

Games for hybrid logic from semantic games to analytic calculi*

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Abstract

Game semantics and winning strategies offer a potential conceptual bridge between semantics and proof systems of logics. We illustrate this link for hybrid logic—an extension of modal logic that allows for explicit reference to worlds within the language. We lift a Hintikka-style semantic game to a *disjunctive game*. The disjunctive game adequately models entailment and validity over classes of frames characterizable in the hybrid language. The search for winning strategies in this game can be reformulated as a sequent-style proof system.

Keywords: hybrid logic, games, proof system

1 Introduction

Games have a long and varied tradition in logic (see, e.g. [1, 13, 18]). Building on the concepts of rational behavior and strategic thinking, they offer a fruitful natural approach to logic, complementing the common paradigm of model-theoretic semantics and proof systems. Game semantics goes back to Jaakko Hintikka [12], who designed a game for two players, called *Me* (or *I*) and *You*,¹ seeking to establish the truth of a formula ϕ of first-order logic in a model $\mathcal{M}$. The game proceeds by rules for step-wise reducing ϕ to an atomic formula. The winning condition depends on the truth of this atomic formula in $\mathcal{M}$. It turns out that *I* have a winning strategy for this game if and only if ϕ is true in $\mathcal{M}$. A natural question is whether there is an algorithm for searching for winning strategies for the game of ϕ over *all* models, which could thus establish (or refute) the validity of ϕ . One such approach is extending the semantic game to a *disjunctive game*. This game can be seen as the two players playing the semantic game overall models simultaneously, thus adequately modelling validity of the involved formula. This technique was first demonstrated in [8] Giles' game for Łukasiewicz logic.

In this paper, we apply this method to the case of hybrid logic. Hybrid logic is an extension of the possible-world semantics of 'orthodox' modal logic, allowing one to explicitly refer to worlds within the object language, using nominals while keeping many attractive features of modal logic intact. Nominals increase expressivity and enable us to grasp many frame properties that are provably not expressible in orthodox modal logic [2]. Apart from this, using nominals can be an advantage for modelling in temporal logic [6] as well as in making the link between modal logic and description logics more explicit [5]. The choice of hybrid logic over orthodox modal logic for developing a game-theoretic approach is natural: direct reference of particular worlds within the language provides conceptual clarity for lifting the semantic game to the disjunctive game.

*This paper is an extension of the conference paper [10].

¹Hintikka and others call this player *Nature*.

2 Games for hybrid logic

The main ingredient for the disjunctive game is allowing *Me* to make ‘backup copies’ of the current game state. *My* aim in the disjunctive game is to win in at least one of these backup copies. Our version of the disjunctive game is the first one to additionally grant *Me* a pool $\mathcal{D}$ of predefined backup states to be used at any time of the game. Logically speaking, these game states could represent frame properties or an underlying theory. The main result of this paper is as follows: *I* have a winning strategy in the disjunctive game if and only if for every model $\mathcal{M}$, *I* have a winning strategy for the initial formula ϕ or some game state in $\mathcal{D}$ in the corresponding semantic game. Although the motivations in this paper are clearly of a logical nature, all the main results and proofs are kept in a game-theoretic language. The corresponding logic facts are straightforward reformulations of the game-theoretic results. Most notably, we show that the disjunctive game adequately models validity and entailment over classes of frames characterizable in the hybrid language, thus proving our version of the disjunctive game a true extension of previous approaches.

The coherence of hybrid logic with games was first demonstrated by Patrick Blackburn, who designed a Lorenzen-style [14] dialogue game for hybrid logic [4]. Sara Negri presented the labelled proof-system **G3K** that resembles the semantics of modal logic [16]. Our approach combines the best of two worlds: it supplements the clear semantic motivation of **G3K** with an accessible game-theoretic viewpoint of Blackburn’s dialogue game. Similar to **G3K**, a failed search for a winning strategy in the disjunctive game directly gives rise to a countermodel. Although the number of players’ possible choices may be infinite in the disjunctive game, we show that winning strategies can be represented as finitely branching trees. This representation allows us to interpret winning strategies as derivations in a proof calculus, similar to other sequent systems for hybrid logic [3, 7].

This paper is structured as follows: Section 2 is a recap of hybrid logic. In Section 3, we present the semantic game and formalize strategies. The disjunctive game over a model is introduced in Section 4 and the general disjunctive game in Section 5. This section also contains the main result. In Section 6, we use this result to directly derive games that adequately model validity and entailment. Finally, we show how winning strategies can be finitized and formulated as proofs in a sequent calculus presented in Section 7.

2 Preliminaries

The language of hybrid modal logic is as follows: we start from two disjoint, countably infinite sets N (set of nominals) and P (set of propositional variables). Nominals are usually called ‘ $i, j, k, \dots$ ’ propositional variables are called ‘ $p, q, \dots$ ’. Formulas ϕ are built according to the following grammar:

$$\phi ::= \perp \mid p \mid i \mid R(i, j) \mid \phi \wedge \phi \mid \phi \vee \phi \mid \phi \rightarrow \phi \mid \neg \phi \mid @_i \phi \mid \Box \phi \mid \Diamond \phi$$

Formulas of the form p, i and $R(i, j)$ are called *elementary*. Intuitively, we can think of the nominal i as the name of a particular world in a model. Hence, i is true in exactly one world. The formula $@_i \phi$ stands for the fact that ϕ is true in the world with the name i . The relational claim $R(i, j)$ says that the world with name j is accessible from the world with name i . Though usually defined as $@_i \Diamond j$, we include it as an elementary formula to prevent circular definitions in the description of the game rules in the following sections.

We now define the semantics formally: a model $\mathcal{M}$ for hybrid modal logic is a tuple (W, R, V, g) , where

1. W is a non-empty set. Its elements are called *worlds*.

2. $R \subseteq W \times W$ is called *accessibility relation*. As usual, we write wRv instead of $(w, v) \in R$. The set of *accessible worlds from w* is $wR := \{v \in W : wRv\}$.
3. $V : P \rightarrow \mathcal{P}(W)$ ² is called *valuation function*.
4. $g : N \rightarrow W$ is called *assignment*. If $g(i) = w$, we say that i is a *name* of w , or simply that w has a name.

The pair (W, R) is called a *frame* and $\mathcal{M}$ is said to be *based on* this frame. Truth of formulas in the world w of W is defined recursively:

$$\begin{aligned}
\mathcal{M}, w &\models \perp \text{ never,} \\
\mathcal{M}, w &\models p, \text{ iff } w \in V(p), \\
\mathcal{M}, w &\models i, \text{ iff } g(i) = w, \\
\mathcal{M}, w &\models R(i, j), \text{ iff } g(i)Rg(j), \\
\mathcal{M}, w &\models \phi \wedge \psi, \text{ iff } \mathcal{M}, w \models \phi \text{ and } \mathcal{M}, w \models \psi, \\
\mathcal{M}, w &\models \phi \vee \psi, \text{ iff } \mathcal{M}, w \models \phi \text{ or } \mathcal{M}, w \models \psi, \\
\mathcal{M}, w &\models \phi \rightarrow \psi, \text{ iff } \mathcal{M}, w \not\models \phi \text{ or }^3 \mathcal{M}, w \models \psi, \\
\mathcal{M}, w &\models \neg\phi, \text{ iff } \mathcal{M}, w \not\models \phi, \\
\mathcal{M}, w &\models @_i\phi, \text{ iff } \mathcal{M}, g(i) \models \phi, \\
\mathcal{M}, w &\models \Box\phi, \text{ iff for all }^4 v \in W, (w, v) \notin R \text{ or } \mathcal{M}, v \models \phi, \\
\mathcal{M}, w &\models \Diamond\phi, \text{ iff for some } v \in W, wRv \text{ and } \mathcal{M}, v \models \phi.
\end{aligned}$$

We say that a formula ϕ is true in the model $\mathcal{M}$ and write $\mathcal{M} \models \phi$ iff $\mathcal{M}, w \models \phi$ for every world w . For a class of models $\mathfrak{M}$, we write $\models_{\mathfrak{M}} \phi$ and say that ϕ is *valid over* $\mathfrak{M}$ iff for all $\mathcal{M} \in \mathfrak{M}$, $\mathcal{M} \models \phi$. We say that ϕ is *valid* (we write $\models \phi$) iff ϕ is valid over the class of all models. For a set of formulas T , we write $\mathcal{M}, w \models T$ iff $\mathcal{M}, w \models \tau$ for every $\tau \in T$. We say that T (*locally*) *entails* ϕ and write $T \models \phi$ iff $\mathcal{M}, w \models T$ implies $\mathcal{M}, w \models \phi$, for every model $\mathcal{M}$ and world w . Similarly, we define $T \models_{\mathfrak{M}} \phi$. We say that $\mathfrak{M}$ is *characterized* by a set of formulas T iff $\mathcal{M} \in \mathfrak{M} \Leftrightarrow \mathcal{M} \models T$, for every model $\mathcal{M}$. If such a T exists, we say that $\mathfrak{M}$ is *characterizable*.

For a frame $\mathcal{F}$, we say that ϕ is *valid over* $\mathcal{F}$ iff $\mathcal{M} \models \phi$, for every model $\mathcal{M}$ based on $\mathcal{F}$. A set of formulas T is *valid over* $\mathcal{F}$ iff $\mathcal{F} \models \tau$ for every $\tau \in T$. A class of frames $\mathfrak{F}$ is *characterized* by a set of formulas T iff $\mathcal{F} \in \mathfrak{F} \Leftrightarrow \mathcal{F} \models T$. In this paper, we will be interested in sets T consisting of *pure* formulas, i.e. not containing propositional variables and closed under nominal substitution (substitution of nominals by nominals). In this case, one can show that $\mathcal{M} \models T$ iff $\mathcal{M}$ is based on a frame from $\mathfrak{F}$. Therefore, we will from now on assume that *characterizable* for a class of frames $\mathfrak{F}$ means characterizable by such an F , and conveniently identify $\mathfrak{F}$ with the class of all models based on a frame from $\mathfrak{F}$ (and use the $\models_{\mathfrak{F}}$ -notation accordingly).

² $\mathcal{P}(W)$ denotes the power set of W .

³This is equivalent to the usual ‘if $\mathcal{M}, w \models \phi$, then $\mathcal{M}, w \models \psi$ ’.

⁴This is equivalent to the usual ‘For all $v \in W$, if wRv , then $\mathcal{M}, v \models \phi$ ’. Our formulation is more easily representable as a game rule.

4 Games for hybrid logic

The most important class for what follows is the class of named models $\mathfrak{N}$. A model is called *named* iff the assignment is surjective, i.e. every world has a name. The accessibility relation in such named models is completely determined by the truth values of the relational formulas. For these models, we can thus state the following semantic facts about the modal operators without explicitly referring to the semantics of the accessibility relation. Let $w = g(i)$, then

$$\begin{aligned}\mathcal{M}, w &\models \Box\phi, \text{ iff for all } j \in N, \mathcal{M}, g(j) \models \neg R(i, j) \vee \phi, \\ \mathcal{M}, w &\models \Diamond\phi, \text{ iff for some } j \in N, \mathcal{M}, g(j) \models R(i, j) \wedge \phi. \\ \mathcal{M} &\models \phi, \text{ iff for all } i \in N, \mathcal{M}, g(i) \models \phi.\end{aligned}$$

We will make use of the fact that questions of validity and entailment can be reduced to named models:

LEMMA 1 (Theorem 7.29 in [2]).

Let T be a countable set of formulas and $\mathfrak{F}$ a characterizable class of frames. Then, for all ϕ ,

$$T \models_{\mathfrak{F}} \phi \iff T \models_{\mathfrak{F} \cap \mathfrak{N}} \phi.$$

3 A game for truth

The *semantic game* is played over a model $\mathcal{M} = (W, R, V, g)$ by two players, *Me* and *You*, who argue about the truth of a formula ϕ at a world w . At each stage of the game, one player acts as proponent, while the other acts as opponent of the claim that a formula ϕ is true at the world w . We represent the situation where *I* am the proponent (and *You* are the opponent) by the *game state* $\mathbf{P}, w : \phi$, and the situation where *I* am the opponent (and *You* are the proponent) by $\mathbf{O}, w : \phi$. We add another kind of game state of the form $\mathbf{P} : \phi$ or $\mathbf{O} : \phi$, representing the claim that ϕ is true in the whole model. We call a game state *elementary* if its involved formula is elementary.

In general, every two-person,⁵ zero-sum, win-lose game is usually represented by a *game-tree*, i.e. a tree whose nodes represent game states and each node is labelled either ‘I’ or ‘Y’. Each run of the game is identified with a path through this game tree: in nodes labelled ‘I’ that have children, it is *My* choice to decide which of the children the game continues at. It is *Your* choice in nodes labelled ‘Y’. Each run of a game ends in a leaf: if it is labelled ‘I’, *I* win and *You* lose if it is labelled ‘Y’, *I* lose and *You* win.

We will now define the game-tree $\mathbf{G}(\mathcal{M}, g)$ for our semantic game starting in the game state g over the model $\mathcal{M}$. The definition is given recursively by specifying the immediate subtrees of $\mathbf{G}(\mathcal{M}, g)$. We state only the cases where *I* am in the role of the proponent. The variants where *I* am in the role of opponent is obtained by switching ‘I’ and ‘Y’ and ‘P’ and ‘O’ in the respective rule.⁶ Note that the rules closely parallel the recursive definition of truth of the previous section:

$$\begin{aligned}(R_{\vee}) \text{ If } g = \mathbf{P}, w : \psi_1 \vee \psi_2, \text{ then } g \text{ is labelled ‘I’ and its immediate subtrees are } \mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi_1) \text{ and } \mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi_2). \\ (R_{\wedge}) \text{ If } g = \mathbf{P}, w : \psi_1 \wedge \psi_2, \text{ then } g \text{ is labelled ‘Y’ and its immediate subtrees are } \mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi_1) \text{ and } \mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi_2).\end{aligned}$$

⁵Let us keep the players’ names *I* and *You*.

⁶For example, for $(R_{\rightarrow})$, if $g = \mathbf{O}, w : \psi_1 \rightarrow \psi_2$, then g is labelled ‘Y’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi_1)$ and $\mathbf{G}(\mathcal{M}, \mathbf{O}, w : \psi_2)$.

- ($R_{\rightarrow}$) If $g = \mathbf{P}, w : \psi_1 \rightarrow \psi_2$, then g is labelled ‘I’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{O}, w : \psi_1)$ and $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi_2)$.
- ($R_{\neg}$) If $g = \mathbf{P}, w : \neg\psi$, then g is labelled ‘I’ and its immediate subtree is $\mathbf{G}(\mathcal{M}, \mathbf{O}, w : \psi)$.
- ($R_{@}$) If $g = \mathbf{P}, w : @_i\psi$, then g is labelled ‘I’ and its immediate subtree is $\mathbf{G}(\mathcal{M}, \mathbf{P}, g(i) : \psi)$.
- ($R_{\Box}$) If $g = \mathbf{P}, w : \Box\psi$ and $wR \neq \emptyset$, then g is labelled ‘Y’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{P}, v : \psi)$, where v ranges over wR . If $wR = \emptyset$, then g is a leaf labelled ‘I’.
- ($R_{\Diamond}$) If $g = \mathbf{P}, w : \Diamond\psi$ and $wR \neq \emptyset$, then g is labelled ‘I’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{P}, v : \psi)$, where v ranges over wR . If $wR = \emptyset$, then g is a leaf labelled ‘Y’.
- ($R_{\perp}$) If $g = \mathbf{P}, w : \perp$, then g is a leaf labelled ‘Y’.
- (R_p) If $g = \mathbf{P}, w : p$, then g is a leaf. It is labelled ‘I’ iff $w \in V(p)$.
- (R_i) If $g = \mathbf{P}, w : i$, then g is a leaf. It is labelled ‘I’ iff $g(i) = w$.
- (R_R) If $g = \mathbf{P}, w : R(i, j)$, then g is a leaf. It is labelled ‘I’ iff $g(i)Rg(j)$.
- (R_U) If $g = \mathbf{P}, w : \psi$, then g is labelled ‘Y’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi)$, where w ranges over W .

We refer to the clauses of this definition as *rules*, because they are rules in the intuitive understanding of the game: for example, the intuitive reading of ($R_{\vee}$) is as follows: if I am currently the proponent of $\psi_1 \vee \psi_2$ at w , then I choose whether to continue the game with ψ_1 or with ψ_2 at w . The rule ($R_{\neg}$) says that at $\neg\psi$, the players change their roles. If I am the proponent of $\Box\phi$ at w , then I immediately win the game if w has no successors. Otherwise, *You* choose a successor v of w , and the game continues with ϕ at v .

Let us show that every run of the game lasts for only finitely many rounds:

PROPOSITION 2.

The tree $\mathbf{G}(\mathcal{M}, g)$ is of finite height.

Proof. We define the degree $\delta(\phi)$ of a formula ϕ to be 0 if ϕ is elementary, $\delta(\phi * \psi) = \max\{\delta(\phi), \delta(\psi)\} + 1$ for $*$ a logical connective and $\delta(\Delta\phi) = \delta(\phi) + 1$ for Δ a modal operator (that includes $@$), or $\neg$. We extend δ to game states by setting $\delta(\mathbf{Q}, w : \phi) = \delta(\phi)$ and $\delta(\mathbf{Q} : \phi) = \delta(\phi) + 1$ for $\mathbf{Q} \in \{\mathbf{P}, \mathbf{O}\}$. The proof now follows from the fact that every rule strictly reduces the degree of the game state. $\square$

EXAMPLE 1.

Consider the model $\mathcal{M}$ in Figure 1 and the following run of the game $\mathbf{G}(\mathcal{M}, \mathbf{P}, w_1 : \Box(j \vee \neg\Box p))$. First, *You* must choose a neighbouring world. Since *You* know that I can defend j at w_2 , let us say that *You* choose w_3 and I must then defend $j \vee \neg\Box p$ at w_3 . Clearly, I will choose the second disjunct. According to the rule of negation, a role switch occurs: I am now the opponent and *You* the proponent of $\Box p$ at w_3 . Hence, I must choose a neighbouring world and *You* must defend p there. As *My* choice is between the p -world w_4 and the non- p -world w_5 , I will choose w_5 and win the game.

This run of the game is represented in Figure 2 and can be found as a path through the game tree in Figure 3. This game tree represents all possible choices of the two players at any point. All leaves of that tree are elementary game states, except for $\mathbf{O}, w_2 : \Box p$, where I immediately win the game because there are no successors of w_2 .

Although not all possible runs of a game end in a winning state for *Me*, I can make choices that will guarantee that the game will end in *My* victory. Generally, if I can always enforce a given game to end in a winning state for *Me*, we say that I have a *winning strategy* for that game. We will make this notion more formal in the following subsection:

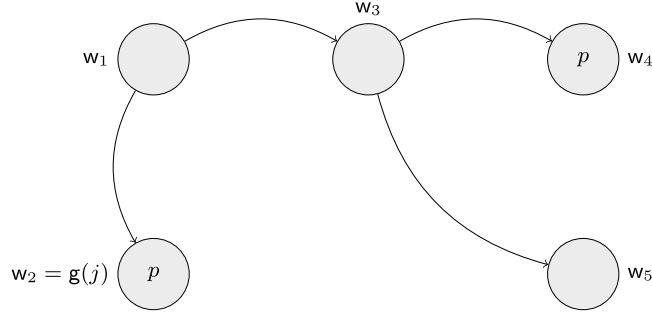


FIGURE 1. Model with worlds $w_1, \dots, w_5$. Arrows represent the accessibility relation. We write p inside the circle representing a world w iff $w \in V(p)$.

$$\begin{array}{c}
 [\mathbf{P}, w_1 : \Box(j \vee \neg\Box p)]^Y \\
 \downarrow \\
 [\mathbf{P}, w_3 : j \vee \neg\Box p]^I \\
 \downarrow \\
 [\mathbf{P}, w_3 : \neg\Box p]^I \\
 \downarrow \\
 [\mathbf{O}, w_3 : \Box p]^I \\
 \downarrow \\
 [\mathbf{O}, w_5 : p]^I
 \end{array}$$

FIGURE 2. A run of the game $\mathbf{G}(\mathcal{M}, \mathbf{P}, w_1 : \Box(j \vee \neg\Box p))$.

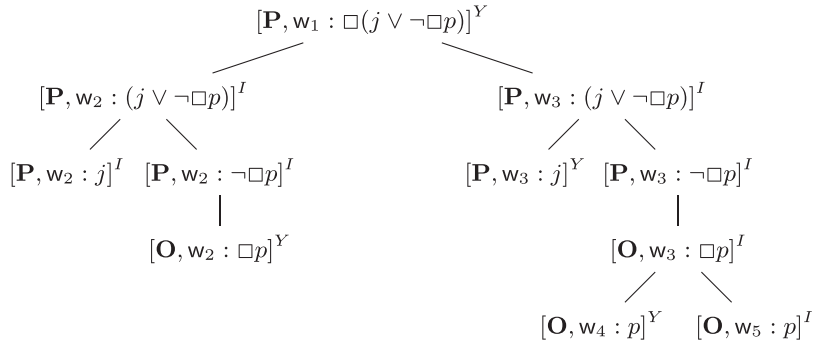
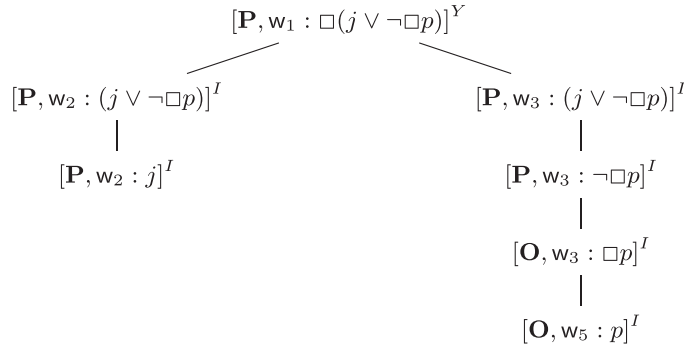


FIGURE 3. The game tree $\mathbf{G}(\mathcal{M}, \mathbf{P}, w_1 : \Box(j \vee \neg\Box p))$.

3.1 Winning Strategies

To precisely describe the scenario where I can enforce that a game ends in a winning state for Me , we need the notion of a winning strategy.

FIGURE 4. A winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w_1 : \Box(j \vee \neg\Box p))$.

DEFINITION 3.

Let T be a game tree. A *strategy*⁷ for *Me/You* for T is a tree σ such that the root of T is in σ and for every node g of T :

- if g is labelled ‘I’/‘Y’, then exactly one successor of g is in σ ,
- if g is labelled ‘Y’/‘I’, then all successors of g are in σ .

A strategy σ is called *winning* for *Me/You* if all leaf-nodes of T in σ are labelled ‘I’/‘Y’.

Intuitively, we can understand a strategy σ for *Me/You* as a game plan: *I/You* play according to σ , or use σ , if at every non-leaf g labelled ‘I’, *I/You* choose to continue the game by the unique successor of g in σ . In Section 4, we will deal with game trees of infinite height. In these games, infinite runs of the game will be considered as winning for *You*. Formally, if T is of infinite height, a strategy for *Me* is winning if it is of finite height, in addition to the above properties. If the tree structure is clear from context (e.g. from the corresponding game tree), we will often identify a strategy with the set of its nodes and write $g \in \sigma$, if g is a node of σ .

EXAMPLE 2.

Continuing the game from Example 1, we can now make precise our observation that *I* can make choices such that the game will always end in winning states for *Me*. A winning strategy for *Me* for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w_1 : \Box(j \vee \neg\Box p))$ is depicted in Figure 4 as a tree.

We will see another example of a game, where *You* have a winning strategy in Example 3 at the end of the section.

Games of finite height are *determined*, i.e. exactly one of the two players has a winning strategy. Let T and S be two game trees. We say that S is the *dual* of T if S and T are isomorphic as trees and the labels of nodes in T are flipped in S . One easily sees that *I* have a winning strategy in the game represented by T iff *You* have a winning strategy in S . We formulate this observation for the semantic game:

⁷In game theory, a strategy for an extensive-form game with perfect information is usually defined as a function from nodes to successors [15]. Since we only care about min-max-strategies, our definition is sufficient (and, for our purposes, more convenient).

LEMMA 4.

For all $\mathcal{M}, w, \phi$ and $\mathbf{Q} \in \{\mathbf{P}, \mathbf{O}\}$, the game $\mathbf{G}(\mathcal{M}, \mathbf{Q}, w : \phi)$ is determined. Furthermore, *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \phi)$ iff *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{O}, w : \phi)$.

This lemma also shows that the existence of a winning strategy for *Me* in $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \phi)$ behaves like a 0/1-truth value. The following theorem shows that there is a complete match between *My* winning strategies and truth values:

THEOREM 5.

Let $\mathcal{M}$ be a model, w a world and ϕ a formula:

1. *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \phi)$ iff $\mathcal{M}, w \models \phi$.
2. *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \phi)$ iff $\mathcal{M}, w \not\models \phi$.
3. *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P} : \phi)$ iff $\mathcal{M} \models \phi$.
4. *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{O} : \phi)$ iff $\mathcal{M} \not\models \phi$.

Proof of Theorem 5. By Lemma 4 and the definition of truth in a model, it suffices to show only 1. We show both directions simultaneously by induction on ϕ . If ϕ is atomic, everything follows from the definition of a winning state. Let us show some of the inductive steps: if ϕ is of the form $\psi \wedge \chi$, then *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi \wedge \chi)$ iff *I* have winning strategies for both $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi)$ and $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \chi)$. By induction hypothesis, this is the case iff $\mathcal{M}, w \models \psi$ and $\mathcal{M}, w \models \chi$, which, in turn, is equivalent to $\mathcal{M}, w \models \psi \wedge \chi$. As for modal operators, *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \Box\psi)$ iff for all successors v of w , *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, v : \psi)$. By the induction hypothesis, this is the case iff $\mathcal{M}, v \models \psi$. Since v is an arbitrary successor of w , this is equivalent to $\mathcal{M}, w \models \Box\psi$. Note that this also covers the trivial case, where there are no successors of w . Lemma 4 is needed for the case of negation: *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \neg\psi)$ iff *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{O}, w : \psi)$. By the lemma, this is the case iff *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, w : \psi)$. By induction hypothesis, this is the case iff $\mathcal{M}, w \not\models \psi$ iff $\mathcal{M}, w \models \neg\psi$. $\square$

We conclude this section with an important observation about the game for named models. Remember that in a named model, every world w has a name i , i.e. $g(i) = w$. Therefore, for $\mathbf{Q} \in \{\mathbf{P}, \mathbf{O}\}$, it is unambiguous if we write $\mathbf{Q}, i : \phi$ for the game state $\mathbf{Q}, w : \phi$. This, together with the fact that $\mathcal{M} \models R(i, j)$ iff $g(i)Rg(j)$ gives us the following equivalent formulations of the rules $(R_\Box)$, $(R_\Diamond)$ and (R_U) :

- $(R_\Box)$ If $g = \mathbf{P}, i : \Box\psi$, then g is labelled ‘Y’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{P}, j : \neg R(i, j) \vee \psi)$, where j ranges over the nominals.
- $(R_\Diamond)$ If $g = \mathbf{P}, i : \Diamond\psi$, then it is labelled ‘I’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{P}, j : R(i, j) \wedge \psi)$, where j ranges over the nominals.
- (R_U) If $g = \mathbf{P} : \psi$, then it is labelled ‘Y’ and its immediate subtrees are $\mathbf{G}(\mathcal{M}, \mathbf{P}, j : \psi)$, where j ranges over the nominals.

To be precise, *equivalent* is to be understood in the following way: two games, T and S are *equivalent* iff *I/You* have a winning strategy for T iff *I/You* have a winning strategy for S . It can be easily seen that the two formulations of the semantic game are equivalent in this sense. In fact, winning strategies can be directly translated and branching over the nominals can be restricted. We formulate this fact as a proposition for reference:

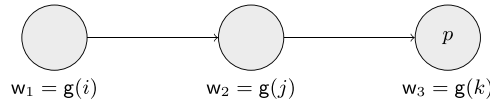
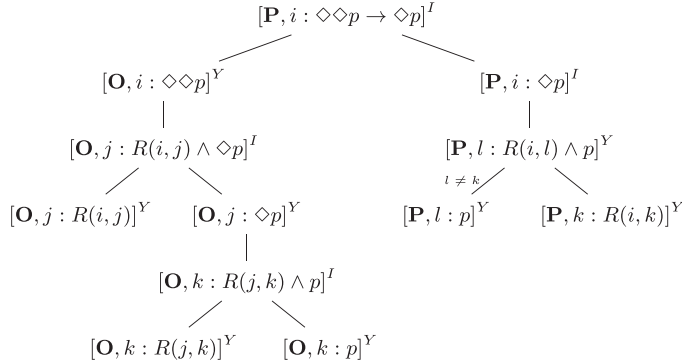


FIGURE 5. A non-transitive model.

FIGURE 6. A winning strategy for *You* for the game $\mathbf{G}(\mathcal{M}, \mathbf{P}, i : \Diamond\Diamond p \rightarrow \Diamond p)$.**PROPOSITION 6.**

Let $\mathcal{M}$ be a model and M a set of nominals. If every world of $\mathcal{M}$ has a name in M , i.e. the restricted function $g : M \rightarrow W$ is already surjective, then for every game state g , the semantic game $\mathbf{G}(\mathcal{M}, g)$ is equivalent to the version with reformulated rules $(R_{\Box})$, $(R_{\Diamond})$ and (R_U) as above, but with nominals ranging over M .

In particular, for such models, Proposition 2,⁸ Lemma 4, and Theorem 5 remain true. Unless noted otherwise, we will from now on use only the reformulated rules from above where j ranges over all nominals. Also, we will tacitly assume that all models are named. This is justified by Lemma 1. We will see the advantages of these reformulations shortly. First, let us consider another example where we use the reformulated rules.

EXAMPLE 3.

Let $\mathcal{M}$ be the model in Figure 5. Because $\mathcal{M}$ is not transitive, we expect *You* to have a winning strategy in the game $\mathbf{G}(\mathcal{M}, \mathbf{P}, i : \Diamond\Diamond p \rightarrow \Diamond p)$. A (compact representation) of *Your* winning strategy is shown in Figure 6. If *I* go to $\mathbf{P}, i : \Diamond p$ in the first turn, *I* will then choose a nominal l and the game continues at $\mathbf{P}, l : R(i, l) \wedge p$. If $l \neq k$, then *You* should go to $\mathbf{P}, l : p$, otherwise to $\mathbf{P}, k : R(i, k)$. In both cases, *You* win the game.

A more direct way of saying that the model is a countermodel to transitivity is by making use of the hybrid language: clearly, $R(i, j) \wedge R(j, k) \rightarrow R(i, k)$ is false at the world $g(i)$. Therefore, *You* have a winning strategy for the game $\mathbf{G}(\mathcal{M}, \mathbf{P}, i : R(i, j) \wedge R(j, k) \rightarrow R(i, k))$.

⁸For the reformulated rules, we need to change the definition of degree: $\delta(\Delta\phi) = \max\{1, \delta(\phi)\} + 1$ for modal operators Δ to deal with the (possibly negated) occurrence of $R(i, j)$ in the successor state.

4 The disjunctive game

This section describes the disjunctive game, an extension of the semantic game. Similarly to the semantic game, the disjunctive game is played over a fixed model $\mathcal{M}$. However, in every game state g , I now have an extra option: instead, of moving according to the rules of the semantic game, I can decide to create a ‘backup-copy’ of g and continue playing at the *disjunctive game state* (or *disjunctive state*) $g \vee g$. If the game is unfavourable for Me in one copy, I can always come back to have another shot at the other copy. Formally, disjunctive game states are finite multisets of game states of the semantic game. We prefer to write $g_1 \vee \dots \vee g_n$ for the disjunctive game state $\{g_1, \dots, g_n\}$, but keep the convenient notation $g \in D$ if g belongs to the multiset set D . A disjunctive state is called *elementary* if all its game states are elementary. My goal is to win in at least one backup copy (hence the ‘disjunctive’). Therefore, disjunctive states are winning for Me if they are elementary and contain at least one winning state of the semantic game for Me .

At a later point, we want to model entailment over theories and frames game-theoretically. To fix ideas, suppose $T \models \phi$. Let $\mathcal{M}$ be a model and i a nominal such that $\mathcal{M}, g(i) \not\models \phi$. By Theorem 5, I do not have a winning strategy in the semantic game over $\mathcal{M}$ starting at $\mathbf{P}, i : \phi$. To capture the fact that $T \models \phi$, we will allow Me to add states of the form $\mathbf{O}, i : \tau$ for $\tau \in T$ as backup-states. In the current example, I will make use of this rule and add some $\mathbf{O}, i : \tau$ with $\mathcal{M}, g(i) \not\models \tau$ and win the game. For now, we want to be general and consider a set $\mathcal{D}$ of game states, called the *context*, as an additional parameter of our game. In the disjunctive game, I can decide to add game states from $\mathcal{D}$ as backup copies.

Note that due to the design of the game, runs of the game can now be infinite. (For example, I can duplicate game states infinitely often). All infinite runs will be considered winning for You .

Another essential technical ingredient is necessary: At a disjunctive game state $g_1 \vee \dots \vee g_n$, it has to be decided which game state is to be played on next. One solution would be to give this choice to Me . However, we prefer to keep things general and introduce *regulation functions*. A regulation function ρ maps non-elementary disjunctive states to non-elementary game states with the property $\rho(D) \in D$. Therefore, if the current disjunctive state is D , the regulation function ‘decides’ that the current game state is $\rho(D)$. In the disjunctive game state $D \vee g$, we will write $D \vee g$ if $\rho(D \vee g) = g$.

We now give the full game-theoretic definition of the disjunctive game specified by the context $\mathcal{D}$, the model $\mathcal{M}$, the regulation function ρ and starting in the disjunctive state D . To this end, we give a recursive definition of its game tree $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ by laying out its immediate subtrees. Here and below, if ρ is clear from the context (or not important), we conveniently write $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D)$.

- If g is labelled ‘Y’ in $\mathbf{G}(\mathcal{M}, g)$,⁹ then so is $D \vee g$. Its immediate subtrees are the games $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D \vee h, \rho)$, where h ranges over all children of g in $\mathbf{G}(\mathcal{M}, g)$. For example, the immediate subtrees of $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D \vee \mathbf{P}, i : \phi \wedge \psi)$ are $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D \vee \mathbf{P}, i : \phi)$ and $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D \vee \mathbf{P}, i : \psi)$.
- If g is labelled ‘I’, then so is $D \vee g$. Its immediate subtrees are the following:
 - $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D \vee h, \rho)$, where h ranges over all children of g in $\mathbf{G}(\mathcal{M}, g)$. We say that ‘ I used a logical rule’,
 - the subtree $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D \vee g \vee g)$, ‘ I duplicated g ’,

⁹More concretely, this means that g is of the form $\mathbf{O}, i : \phi \vee \psi$, $\mathbf{P}, i : \phi \wedge \psi$, $\mathbf{O}, i : \phi \rightarrow \psi$, $\mathbf{O}, i : @_j \phi$, $\mathbf{P}, i : \Box \phi$, $\mathbf{O}, i : \Diamond \phi$, or $\mathbf{P} : \phi$.

- $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D \vee g \vee h)$, where h ranges over the game states in $\mathcal{D}$, ‘ I used a context rule’,
- The subtree $\mathbf{DG}^{\emptyset}(\mathcal{M}, D \vee g)$.
- If D is elementary and $\mathcal{D} \neq \emptyset$, then it is labelled ‘I’ and its immediate subtree is $\mathbf{DG}^{\emptyset}(\mathcal{M}, D)$. If $\mathcal{D} = \emptyset$, then it is a leaf. It is labelled ‘I’, if some $g \in D$ is labelled ‘I’ in $\mathbf{G}(\mathcal{M}, g)$. Otherwise, it is labelled ‘Y’.

Going from $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D)$ with $\mathcal{D} \neq \emptyset$ to $\mathbf{DG}^{\emptyset}(\mathcal{M}, D)$, we say that ‘ I deleted the context’. Note that a run of the game can only reach a leaf if the context is empty, i.e. has been empty from the beginning or has been deleted by *Me*. This way, we make sure that in every run of the game, *I* have the option to use the context rule.

REMARK 1.

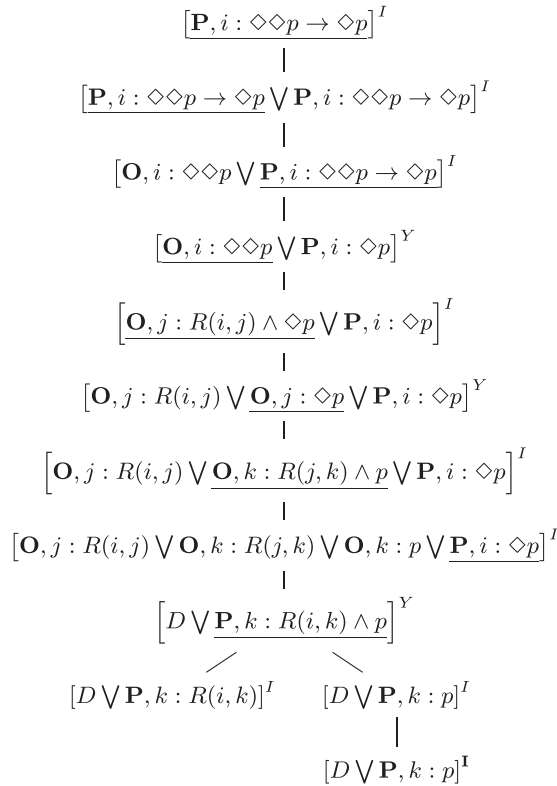
(1) Game-theoretically, the disjunctive game is again a game. (Winning) strategies for the disjunctive game are therefore well-defined by Definition 3. It even makes sense to talk about ‘game states’ of the disjunctive game. However, to prevent confusion, we will reserve this term for the game states of the semantic game only and use ‘disjunctive game state’ when we talk about the disjunctive game.

(2) Due to the rules of duplication and context, runs of the game can now be infinite. Remember that infinite runs of the game are considered winning for *You*. However, our games retain a vital property: by the Gale–Stewart Theorem [11], every instance of the disjunctive game is *determined*, i.e. exactly one of the two players has a winning strategy. A direct proof follows from the two propositions below.

(3) Since the regulation ρ is a function from non-elementary disjunctive states D to non-elementary game states $g \in D$, it does not depend on the history of the current run of the game, i.e. the sequence of disjunctive states that led up to D . Therefore, ρ is ‘forced’ to always pick the same game state at D , independent of previous moves. This memoryless behavior may seem an unnatural restriction on the generality of ρ . To remedy this, we could change our definition of ρ to map finite sequences of disjunctive states $\langle D_0, \dots, D_n \rangle$ to $g \in D_n$, or formalize ρ as a non-strategic player of the disjunctive game. Since all following proofs from here on can easily be modified to match these definitions, we stick to the technically easier original version.

EXAMPLE 4.

In Example 2, we have seen the semantic game $\mathbf{G}(\mathcal{M}, \mathbf{P}, w_1 : \Box(j \vee \neg\Box p))$, where *I* have a winning strategy. It therefore makes no sense for *Me* to use the duplication rule in the corresponding disjunctive game $\mathbf{DG}^{\emptyset}(\mathcal{M}, \mathbf{P}, w_1 : \Box(j \vee \neg\Box p), \rho)$. In Example 3, *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, i : \Diamond\Diamond p \rightarrow \Diamond p)$. In the disjunctive game $\mathbf{DG}^{\emptyset}(\mathcal{M}, \mathbf{P}, i : \Diamond\Diamond p \rightarrow p, \rho)$, using the duplication rule only produces backup copies of game states, where *You* have a winning strategy and therefore does not help *Me*. In fact, we will see in the following results that the duplication rule does not change *My* and *Your* winning powers if the context $\mathcal{D}$ is empty. Finally, let us look at a case where the context is not empty. With $\mathcal{M}$ as in Example 3, consider the game $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, \mathbf{P}, i : \Diamond\Diamond p \rightarrow p, \rho)$, where $\mathcal{D} = \{\mathbf{O}, i : R(i, j) \wedge R(j, k) \rightarrow R(i, k) \mid i, j, k \in N\}$. In this game, *I* can make use of the context rule to add $\mathbf{O}, i : R(i, j) \wedge R(j, k) \rightarrow R(i, k)$ as a backup-copy. Since *I* have a winning strategy for this game state in the semantic game, *I* will win the disjunctive game. This illustrates the fact that for *Me* to win the disjunctive game, it is enough to have a winning strategy for some game state in the disjunctive state or in the context. This fact is also formally proved below. Furthermore, *My* winning

FIGURE 7. A winning strategy for the game $\mathbf{DG}^D(\mathcal{M}, \mathbf{P}, i : \Diamond\Diamond p \rightarrow \Diamond p, \rho)$.

strategy for this game can be seen as a witness of the semantic fact

$$R(i, j) \wedge R(j, k) \rightarrow R(i, k) \models @_i(\Diamond\Diamond p \Rightarrow \Diamond p).$$

A depiction of *My* winning strategy for this game can be found in Figure 7, where as usual, the marked game state is chosen by the regulation function ρ . In the state $\mathbf{O}, i : \Diamond\Diamond p \vee \mathbf{P}, i : \Diamond p$, we avoided writing down the other options for *You*, since they easily result in a win for *Me*. In the state $\mathbf{O}, j : R(i, j) \wedge \Diamond p \vee \mathbf{P}, i : \Diamond p$, I used the duplication rule, and went to $\mathbf{O}, j : R(i, j)$ in one, and to $\mathbf{O}, j : \Diamond p$ in the other copy. We do not mention the other applications of this trick. D is an abbreviation for the elementary state $\mathbf{O}, j : R(i, j) \vee \mathbf{O}, k : R(j, k) \vee \mathbf{O}, k : p$. The tree continues at $D \vee \mathbf{P}, k : R(i, k)$, abbreviated as H in Figure 8 with an application of the context rule. In elementary states, we write the label ‘I’/‘Y’ bold to indicate that *I* deleted the context.

This winning strategy may seem a bit overblown since it includes many non-essential moves. However, we will see that it contains the necessary information to win the disjunctive game over *all* models. Indeed, the leaves are not only true in the model $\mathcal{M}$ but valid over all models. Therefore, this winning strategy can even be seen as *proof* for the above semantic fact. Strategies like these will be the topic of the next section.

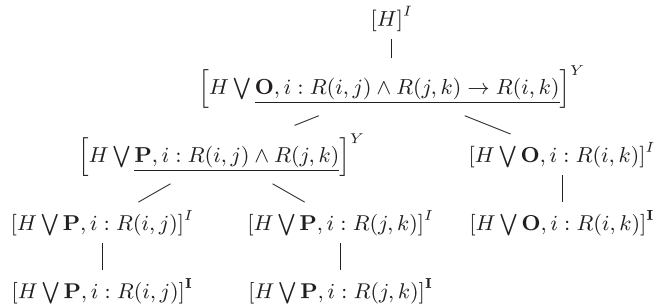


FIGURE 8. A winning strategy for $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, H)$, where $\mathcal{D}$ characterizes transitive frames.

We now compare the disjunctive game to the semantic game from a strategic viewpoint. Essentially, a strategy for *Me* for the disjunctive game over a model is nothing more but a disjunction of strategies for the semantic game:

PROPOSITION 7.

I have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ iff *I* have a winning strategy in $\mathbf{G}(\mathcal{M}, g)$ for some $g \in D \cup \mathcal{D}$.

Proof. ‘ $\Rightarrow$ ’: let σ be a winning strategy for *Me* for $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$. We proceed by bottom-up tree recursion on σ to define for each disjunctive state $H \in \sigma$ a strategy σ_H . The invariant of this construction is the fact that σ_H is a winning strategy for *Me* for the game $\mathbf{G}(\mathcal{M}, h)$ for some $h \in H \cup \mathcal{D}$. The proposition follows from the fact that σ is rooted in D .

If H is a leaf, then, by definition, at least one $h \in H$ must be winning for *Me*. We can therefore set $\sigma_H = \{h\}$.

If $H = K \vee \underline{k}$ is labelled ‘Y’, then all its immediate successors are of the form $K \vee k'$, where k' ranges over the children of k in $\mathbf{G}(\mathcal{M}, k)$. Suppose we have already defined the strategies $\sigma_{K \vee k'}$ and they satisfy the invariant. If some of these strategies is a winning strategy for an $h \in K \cup \mathcal{D}$, then, in particular, it is a winning strategy for some $h \in H \cup \mathcal{D}$, which means we can set $\mu_H = \mu_{K \vee k'}$. Otherwise, every $\sigma_{K \vee k'}$ is a winning strategy for $\mathbf{G}(\mathcal{M}, k')$. In this case, we can set $\mu_H = \{k\} \cup \bigcup_{k'} \mu_{K \vee k'}$, which will be a winning strategy for $\mathbf{G}(\mathcal{M}, k)$: in the first turn, *You* go to k' and *I* continue playing according to *My* winning strategy $\mu_{K \vee k'}$.

If $H = K \vee \underline{k}$ is labelled ‘I’, then there is exactly one child L of H in σ . If $L = K \vee k \vee k$ (*I* used the duplication rule), or $L = K \vee k \vee d$, for some $d \in \mathcal{D}$ (*I* used the context rule), or $L = K \vee k$ (*I* deleted the context), then we are in a trivial case: by hypothesis, μ_L is a winning strategy for some state in $K \cup \{k\} \vee \{k\} \cup \mathcal{D}$, $K \cup \{k\} \cup \{d\} \cup \mathcal{D}$, or $K \cup \{k\} \cup \mathcal{D}$, respectively. All these sets equal $H \cup \mathcal{D}$, so we can simply set $\mu_H = \mu_L$. If *I* used a logical rule, i.e. $L = K \vee k'$ for some child k' of k in $\mathbf{G}(\mathcal{M}, k)$ and μ_L is a winning strategy for some $K \cup \mathcal{D}$, then, in particular, it is a winning strategy for some state in $K \cup \{k\} \cup \mathcal{D}$, so we again set $\mu_H = \mu_L$. If μ_L is a winning strategy for k' , we set $\mu_H = \{k\} \cup \mu_L$, which makes it a winning strategy for *Me* in $\mathbf{G}(\mathcal{M}, k)$: in the first round *I* go to k' and then continue playing according to μ_L .

‘ $\Leftarrow$ ’: suppose, *I* have a winning strategy σ for $\mathbf{G}(\mathcal{M}, g)$ for some $g \in D \cup \mathcal{D}$. First note that in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$, *I* have a strategy to reach a disjunctive state H_0 containing g : This is trivial, if g already appears in D . Otherwise, $g \in \mathcal{D}$, so the context is non-empty. Therefore, *I* can add g at any ‘I’-node, at the latest, at an elementary disjunctive state. My strategy starting from H_0 is as follows: if the current disjunctive state is $H \vee \underline{h}$, where h appears in σ and has label ‘I’, then it has exactly one

successor h' in σ . Consequently, I move to $H \vee h'$. If h does not appear in σ , then I move arbitrarily using the corresponding logical rule. Playing this way I ensure two things: Since I do not make use of duplication and context rules, the game reaches an elementary disjunctive state after finitely many rounds. Secondly, the disjunctive game state always contains a game state from σ . This can be seen using an easy tree induction. Upon reaching an elementary disjunctive state, I delete the context if necessary. The leaf H contains a game state from σ , which is winning for Me . Hence, I win the game at H . $\square$

REMARK 2.

In particular, this proposition shows that regulations play no big role in the disjunctive game over a model. In the next section, the players will play the disjunctive game without any information about the underlying model. We will see that in this case, a particular class of ill-behaved regulation functions can hinder Me from winning games in which we would otherwise expect Me to have a winning strategy.

Another characterization in terms of *Your* strategies shows that a winning strategy for *You* in the disjunctive game is a conjunction of strategies for the semantic game:

PROPOSITION 8.

You have a winning strategy in $\mathbf{DG}^D(\mathcal{M}, D, \rho)$ iff *You* have winning strategies in $\mathbf{G}(\mathcal{M}, g)$ for all $g \in D \cup \mathcal{D}$.

This fact follows immediately from the Gale–Stewart Theorem and Proposition 7. However, there is also a direct constructive proof:

Proof of Proposition 8. ‘ $\Rightarrow$ ’: let σ be a winning strategy for *You* in the game $\mathbf{DG}^D(\mathcal{M}, D, \rho)$ and $g \in \mathcal{D} \cup D$. Since σ is winning, it must contain a disjunctive state of the form $H = H' \vee g$: if g is in D , then H is simply D . If g is in $\mathcal{D}$, then I can add g at any disjunctive state labelled ‘I’, for example, at elementary disjunctive states. Let T be the tree rooted in H , where I make use only of logical rules or delete the context and *You* play according to σ . Note that T is necessarily of finite height, and all its nodes are contained in σ . We will now see that T contains all necessary information for *Your* winning strategies for the required semantic games.

By bottom-up recursion on T , we define a winning strategy μ_h for *You* in $\mathbf{DG}(\mathcal{M}, h)$ for every $K \in T$ and $h \in K$. If K is elementary, then by assumption on σ , for all $h \in K$, $\mathbf{G}(\mathcal{M}, h)$ must be winning for *You*, so set $\mu_h = \{h\}$. If $K = K' \vee \underline{k}$ is a ‘Y’-node, it has exactly one successor $K' \vee k'$. For the recursion, we suppose that we already have a winning strategy for all states in $K \vee k'$, so we only need to define a winning strategy for $\mathbf{G}(\mathcal{M}, k)$. Let $\mu_{k'}$ be the winning strategy for $\mathbf{G}(\mathcal{M}, k')$, then we can set $\mu_k = \{k\} \cup \mu_{k'}$. Finally, if $K' \vee \underline{k}$ is an ‘I’-node, then its successors are $K' \vee k'$, where the k' are the children of k in $\mathbf{G}(\mathcal{M}, k)$. Note that we excluded the case where I use another rule in the construction of T . By our recursive construction, μ_h is winning for all $h \in K'$ and the $\mu_{k'}$ are winning for the k' s. Therefore, set $\mu_k = \{k\} \cup \bigcup_{k'} \mu_{k'}$.

By the inductive hypothesis, we have winning strategies for all $h \in K'$ and a winning strategy $\mu_{k'}$ for $\mathbf{G}(\mathcal{M}, h)$.

By hypothesis, μ_h is winning for all $\mathbf{G}(\mathcal{M}, h)$ for $h \in K'$ and $h = k'$. We do not have a winning strategy for $\mathbf{G}(\mathcal{M}, k)$ yet, but we can set $\sigma_k = \{k\} \cup \sigma_{k'}$.

For h' , we must set $\sigma_{h'} = \{h'\} \cup \sigma_{h''}$. Finally, if $K' \vee h'$ is an ‘I’-node, then its successors are $K' \vee h''$, where the h'' are the children of h' in $\mathbf{G}(\mathcal{M}, h')$. By inductive hypothesis, μ_h is winning for all $h \in K'$ and the $\mu_{h''}$ are winning for the h'' s. Therefore, set $\mu_{h'} = \{h'\} \cup \bigcup_{h''} \mu_{h''}$. ‘ $\Leftarrow$ ’: let σ_g be a winning strategy for *You* for each $g \in D \cup \mathcal{D}$. *Your* strategy for $\mathbf{DG}^D(\mathcal{M}, D, \rho)$ is as follows: if

the current disjunctive state is $H \vee h$, h is labelled ‘Y’ and appears in σ_g for some $h \in \sigma_g$, then it has exactly one successor h' in σ_g . Now, *You* move to $H \vee h'$. Playing this way, all disjunctive states will only consist of game states from the σ_g s. This is shown by a simple root-down induction: the important step is that if *I* make use of the context rule and add a game state g to the current disjunctive state, *You* have a winning strategy σ_g prepared. Every run of the game is either infinite or ends at an elementary disjunctive state. In the first case, *You* win by definition. As all game states of elementary disjunctive states are winning for *You*, *You* win the game in the second case, too. $\square$

5 Forgetting the model

In this section, we look at the game $\mathbf{DG}^{\mathcal{D}}(D, \rho)$. Intuitively, it can be interpreted as the scenario where the players of the disjunctive game $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ forgot—or have not been informed—about the structure of the model $\mathcal{M}$. The goal of both players is to come up with strategies that guarantee them a win, independent of what the model $\mathcal{M}$ looks like.

Let us turn to the formal definition of $\mathbf{DG}^{\mathcal{D}}(D, \rho)$. The definition of the game tree is identical to that of $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$, except for elementary disjunctive states¹⁰:

- If D is elementary and $\mathcal{D} \neq \emptyset$, then it is labelled ‘I’. If $\mathcal{D} = \emptyset$, then it is a leaf. It is labelled ‘I’, if for all models $\mathcal{M}$, D is labelled ‘I’ in $\mathcal{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$. Otherwise, it is labelled ‘Y’.

Why does this simple formal alteration of the rules capture our intuitions of the two players having to play the game over the model ‘blindly’? The reason lies in a strong uniformity of game trees of the disjunctive game. To be precise, the game trees of the disjunctive games $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}_1, D, \rho)$ and $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}_2, D, \rho)$ are identical, except maybe for the labelling of leafs. That is the reason why we can use the game tree $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ (for an arbitrary $\mathcal{M}$) for the game $\mathbf{DG}^{\mathcal{D}}(D, \rho)$ and have to change only the leaves. Note that this uniformity was achieved essentially by our change of the rules (R_U) , $(R_{\square})$ and $(R_{\Diamond})$, i.e. the use of the hybrid language.

EXAMPLE 5.

Let $\mathcal{D} = \{\mathbf{O}, i : R(i, j) \wedge R(j, k) \rightarrow R(i, k) \mid j, k \in N\}$. Figure 7 depicts a winning strategy for *Me* in the game $\mathbf{DG}^{\mathcal{D}}(\mathbf{P}, i : \Diamond p \rightarrow \Diamond p, \rho)$ with ρ indicated as usual. Now, in the states where *You* choose a nominal, let j be *Your* choice for $\Diamond p$ and k *Your* choice for $\Diamond p$.¹¹ The leafs are (from left to right): $\mathbf{O}, j : R(i, j) \vee \dots \vee \mathbf{P}, i : R(i, j)$, $\mathbf{O}, k : R(j, k) \vee \dots \vee \mathbf{P}, i : R(j, k)$ and $\mathbf{O}, k : p \vee \dots \vee \mathbf{P}, k : p$ and indeed winning for *Me* in all models.

Intuitively, a winning strategy for $\mathbf{DG}^{\mathcal{D}}(D, \rho)$ should contain the information how to win $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ for every model $\mathcal{M}$. Indeed, this is true for *My* winning strategies:

THEOREM 9.

If σ is a winning strategy for *Me* in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$, then σ is also a winning strategy for *Me* in $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ for every model $\mathcal{M}$.

Proof. The game trees of the two games are identical, except maybe for the labels of the leaves: But all leafs labelled ‘I’ in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$ have the same label in $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$. $\square$

¹⁰and writing ‘ $\mathbf{DG}^{\mathcal{D}}(D, \rho)$ ’ instead of ‘ $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ ’

¹¹Technically, we need to include branching over all nominals, but the corresponding game trees are uniform in the choice of nominals. In Section 7.1 we will see that choosing nominals not appearing in the disjunctive state is—in a way—an *optimal* move for *You*. The game tree can therefore be finitized in the way that we did.

On the level of existence, the contraposition of the theorem is ‘If *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ for one model $\mathcal{M}$, then *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$ ’. Using a similar proof, however, one can show the following constructive result:

THEOREM 10.

If σ is a winning strategy for *You* in $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$ for one model $\mathcal{M}$, then it is also a winning strategy for *You* in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$.

Next, we want to show that *Your* winning strategy in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$ gives rise to a model $\mathcal{M}$ and a winning strategy for *You* in $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$. However, this is not true for general regulations since some do not allow for a ‘level playing field’ in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$. In these games, winning chances are skewed such that *You* can win even without having a countermodel.

EXAMPLE 6.

Consider a ‘nasty’ regulation ρ , that always chooses an ‘I’-state, if there is one. By Remark 2, this does not affect *Me* in the game $\mathbf{DG}^{\emptyset}(\mathcal{M}, \mathbf{P} : p \vee \mathbf{O} : p, \rho)$. In fact, Proposition 7 and Theorem 5 imply that *I* have a winning strategy for every model $\mathcal{M}$.

The situation changes, when we consider the game $\mathbf{DG}^{\emptyset}(\mathbf{P} : p \vee \mathbf{O} : p, \rho)$. In this game *You* have a winning strategy: due to the nature of ρ , there are two possible kinds of runs of the game. Either throughout the run, there is always a copy of $\mathbf{O} : p$ in the disjunctive game state. Since $\mathbf{O} : p$ is an ‘I’-state, this run is infinite, and *You* win by definition. Otherwise, at some point, the game reaches a disjunctive state

$$\mathbf{P} : p \vee \mathbf{O}, i_1 : p \vee \dots \vee \mathbf{O}, i_n : p,$$

where it is finally *Your* turn. *You* win the game by choosing a nominal j different from all of $i_1, \dots, i_n$ since there is a simple two-world-model where all the above game states are winning for *You*.

In contrast to such ill-behaved regulations, let us introduce the notion of a *fair* regulation. First, we need the following graph-theoretic notion: let T be a tree and B a set of nodes. B is called *bar* in T if every infinite path from the root of T eventually passes through B . A regulation ρ is called *fair* for D if $\{H : \rho(H) = g\}$ is a bar in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$, for all non-elementary $g \in D$. It is called *fair*, if it is fair for all disjunctive game states. Intuitively, this means that in every disjunctive state containing a certain non-elementary game state, this state will be picked eventually, no matter how the players behave. The regulation from the above example is not fair since there is a run where the state $\mathbf{P} : p$ is never reached.

THEOREM 11.

Let ρ be fair. If *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D, \rho)$, then there is a model $\mathcal{M}$ such that *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}, D, \rho)$.

The proof will be the subject of the following subsection. It uses techniques from [3] and [16].

5.1 My best way to play

We will now describe a strategy σ for *Me* for the game $\mathbf{DG}^{\mathcal{D}}(D_0, \rho)$. Intuitively σ will exploit all of *My* possible choices without sacrificing *My* winning chances. Consequently, σ is winning iff *I* can win the game at all.¹²

¹²We will not prove this claim, but it follows from Theorem 11 by classical reasoning.

Throughout the game I will keep track of a number $\#g$ for every game state g and the number n of My turns. At the beginning D_0 , we set $n = 1$ and $\#g = 1$ for every g . Furthermore, suppose the children of g in $\mathbf{G}(g)$ are enumerated: $g_1, g_2, \dots$ and so is $\mathcal{D} = \{h_1, h_2, \dots\}$. Let $|g|$ denote the number of children of g in the semantic game (remember that this is independent of the model $\mathcal{M}$). For a disjunctive state D , we denote by D^{el} the elementary part of D . We call a disjunctive state D *winning* if D^{el} is labelled ‘I’ in $\mathbf{DG}^\emptyset(D^{el})$ (i.e. it is a winning leaf for Me).

The strategy σ is as follows: let D be a non-leaf labelled ‘I’:

1. If D is not winning and it is My n th turn,
 - a. if n is even and $n/2 \leq |\mathcal{D}|$, then move to $D \vee h_{n/2}$.
 - b. if n is even and $n/2 = |\mathcal{D}| + 1$, then delete the context,
 - c. if n is even and $n/2 > |\mathcal{D}| + 1$, proceed to $n + 1$.
 - d. if n is odd, $D = D' \vee \underline{g}$ is not elementary, then augment $\#g$ by 1 after the following step:
 - i. Move to $D' \vee g_{\#g/2}$, if $\#g$ is even and $\#g \leq 2|g|$,
 - ii. move to $D' \vee g \vee \underline{g}$, if $\#g$ is odd and $\#g < 2|g| - 1$,
 - iii. otherwise, move to an arbitrary child $D' \vee g'$.
2. If D is winning,
 - a. $D = D' \vee \underline{g}$ is not elementary, then use a logical rule and move to an arbitrary child $D' \vee g'$.
 - b. D is elementary, then delete the context.

To put it in words: until the game reaches a winning disjunctive state, My strategy is to alternate between logical rules and context rules in such a way that a) I always duplicate a state, then play by exhausting all possible applications of logical rules to that state one after the other and b) include all game states from the context, one after the other. If the game reaches a winning disjunctive state, I terminate the game as quickly as possible. The only fact from this construction that we need is the following: let π be the path through $\mathbf{DG}^\mathcal{D}(D_0, \rho)$ resulting from You playing according to $Your$ winning strategy and Me playing My best way σ . We say that a game state g *appears along* π , and write $g \in \pi$ if it occurs as part of a disjunctive state in π . We have the lemma:

LEMMA 12.

Let π be as above. Then

- 1) Every game state of $\mathcal{D}$ appears in π .
- 2) Let $g \in \pi$ be a non-elementary game state labelled ‘Y’ in the semantic game. Then at least one successor of g appears along π .
- 3) Let $g \in \pi$ be a non-elementary game state labelled ‘I’ in the semantic game. Then all successors of g appear in π .

Proof. First, note that since You play according to $Your$ winning strategy, π never reaches a winning disjunctive state for Me , i.e. Case 2 in the definition of σ .

1) Let $g \in \mathcal{D}$ be h_m . Since $\mathcal{D}$ is not empty and we never reach a winning disjunctive state, the game cannot end until I delete the context by definition of the game. This will not happen before My $(2m + 2)$ -nd turn (1b). When the game reaches My $2m$ th turn, I use the context rule to add h_m (1a). 2) Let $D \vee g$ be a disjunctive state in π , where g is non-elementary and labelled ‘Y’. There are two cases: if π is finite, then at some point it will reach a state of the form $H \vee \underline{g}$, otherwise the game cannot end. If π is infinite, then it will reach such a state by the fairness condition on ρ . In any case, You play according to $Your$ strategy, and the game reaches $H \vee g'$, where g' is a child of g . 3) Let g be labelled ‘I’ and let $D \vee g$ be the earliest appearance of g in π , i.e. $\#g = 1$. The game cannot end as long as the current disjunctive states contains a copy of g . By an easy induction we

can show the following facts: First, the current disjunctive state will always contain a copy of g , as long as $\#g < 2|g|$ (by 1d, ii. I always duplicate g , before I use it up and move to one of its children). Secondly, by fairness of ρ , as long as $\#g \leq 2|g|$, the game will always reach a disjunctive state where g is chosen by the regulation. This implies that $\#g$ will strictly increase throughout the game, and for every $m \leq 2|g|$ reach $2m$, where, by 1d, i., g_m is added. $\square$

We can now show that π gives rise to a model $\mathcal{M}_\pi$ with the property that *You* have a winning strategy for every g appearing along π . In particular, this gives *You* a winning strategy for $\mathbf{DG}^D(\mathcal{M}_\pi, D_0, \rho)$ by Proposition 8. Since we will need it in a later section, we give the following definition in a bit more generality:

DEFINITION 13.

Let $\mathcal{E}$ be a set of elementary game states. We define the relation $i \sim_{\mathcal{E}} j$ between two nominals i and j iff $\mathbf{O}, i : j$ or $\mathbf{O}, j : i$ appear in $\mathcal{E}$. Let $\approx_{\mathcal{E}}$ be the symmetric, reflexive and transitive closure of $\sim_{\mathcal{E}}$. We write $[i]$ for the equivalence class of i . Let $\mathcal{M}_{\mathcal{E}}$ be the following named model:

- Worlds W : equivalence classes of nominals,
- Accessibility relation $R_{\mathcal{E}}$: we have $[i]R_{\mathcal{E}}[j]$ iff for some $i' \in [i], j' \in [j], k \in N$, the state $\mathbf{O}, k : R(i', j')$ is in $\mathcal{E}$.
- Valuation function $V_{\mathcal{E}}$: $[i] \in V_{\mathcal{E}}(p)$ iff for some $i' \in [i]$, the state $\mathbf{O}, i' : p$ is in $\mathcal{E}$.
- Assignment $g_{\mathcal{E}}$: $g_{\mathcal{E}}(i) = [i]$.

We are interested in the case where $\mathcal{E}$ is the set of all elementary game states appearing along π and write $\mathcal{M}_\pi$ instead of $\mathcal{M}_{\mathcal{E}}$. Let us look at a simple example. Let $\mathcal{D} = \{\mathbf{O}, k : R(k, k) \mid k \in N\}$ and consider the game $\mathbf{DG}^D(\mathbf{O}, i : j)$. *My* best way to play is to keep including game states $\mathbf{O}, k : R(k, k)$. This procedure never leads to an elementary winning disjunctive state for *Me*, but results in a single infinite path π , containing all game states from $\mathcal{D}$ as well as the starting state $\mathbf{O}, i : j$. The model $\mathcal{M}_\pi$ looks as follows: Its worlds are the equivalence classes $[k]$, where $[k_1] = [k_2]$ if and only if k_1 and k_2 are equal or if one is i and the other j . For every world $[k]$, we have $[k]R_\pi[k]$ (and no other relations). The resulting model is thus reflexive.

LEMMA 14.

Let $\mathcal{M}_\pi$ be the model $\mathcal{M}_{\mathcal{E}}$ from the above definition, where $\mathcal{E}$ is the set of all elementary game states appearing along π . If g appears along π , then *You* have a winning strategy for $\mathbf{G}(\mathcal{M}_\pi, g)$.

Proof. We prove this lemma by induction on the degree of g . The elementary cases where g is of the form $\mathbf{O}, i : \phi$ are follow directly from the definition of $\mathcal{M}_\pi$. Assume $g = \mathbf{P}, i : p$ appears along π , but $\mathcal{M}_\pi, [i] \models p$. The latter implies that for some $j \in [i]$, $\mathbf{O}, j : p$ appears along π . Since the elementary states of π are accumulative, there is a disjunctive state D in π containing the three states $\mathbf{P}, i : p$, $\mathbf{O}, j : p$ and one of $\mathbf{O}, i : j$ or $\mathbf{O}, j : i$. Clearly, there is no model $\mathcal{M}$ satisfying $\mathcal{M}, g(i) \not\models p$, $\mathcal{M}, g(j) \models p$ and $g(i) = g(j)$ at the same time. Thus, I could have easily won the game starting at D , a contradiction to the fact that σ is not a winning strategy for *Me* in $\mathbf{DG}(D, \rho)$. The cases for $\mathbf{P}, k : R(i, j)$ and $\mathbf{P}, i : j$ are similar.

For the inductive step, let $g \in \pi$ be non-elementary with label ‘Y’. By Lemma 12, some child g' of g appears along π . By inductive hypothesis, there is a winning strategy $\mu_{g'}$ for *You* for $\mathbf{G}(\mathcal{M}_\pi, g')$. Thus, $\mu_g = \{g\} \cup \mu_{g'}$ is a winning strategy for $\mathbf{G}(\mathcal{M}_\pi, g)$.

If g is non-elementary with label ‘I’, then, by Lemma 12, all children g' of g appear along π . For each g' there is a winning strategy for *You* in $\mathbf{G}(\mathcal{M}_\pi, g')$. Thus, $\mu_g = \{g\} \cup \bigcup_{g'} \mu_{g'}$ is a winning strategy for *You* for $\mathbf{G}(\mathcal{M}_\pi, g)$. $\square$

Proof of Theorem 11. Suppose, *You* have a winning strategy for the game $\mathbf{DG}^{\mathcal{D}}(D_0, \rho)$. Let *Me* play according to the strategy σ from above and let π be the resulting path through $\mathbf{DG}^{\mathcal{D}}(D_0, \rho)$ and let $\mathcal{M}_\pi$ be the model form Definition 13. By Lemma 12 and Lemma 14, *You* have a winning strategy for $\mathbf{G}(\mathcal{M}_\pi, g)$ for all $g \in D_0 \cup \mathcal{D}$. Therefore, by Proposition 8, *You* have a winning strategy for $\mathbf{DG}^{\mathcal{D}}(\mathcal{M}_\pi, D_0, \rho)$. $\square$

6 Games for hybrid logic

In this section, we use our results so far to present several games for hybrid logic in terms of the disjunctive game from the previous subsection. For each game, we state the corresponding adequacy result. Throughout the section, let ρ be a fair regulation. Remember that we assume that all models are named.

1. **A game for validity:** let ϕ be (a formula).

$$I \text{ have a winning strategy for } \mathbf{DG}(\mathbf{P} : \phi, \rho) \iff \phi \text{ is valid.}$$

2. **A game for entailment:** let T be a set of formulas and set $\mathcal{D}_i(T) = \{\mathbf{O}, i : \tau \mid \tau \in T\}$. Let $\mathbf{E}(T, \phi, \rho)$ be the following game: in the first round, *You* choose a nominal i . The game then continues with $\mathbf{DG}^{\mathcal{D}_i(T)}(\mathbf{P}, i : \phi, \rho)$.

$$I \text{ have a winning strategy for } \mathbf{E}(T, \phi, \rho) \iff T \models \phi.$$

3. **A game for global consequence:** we say that T globally entails ϕ and write $T \models_g \phi$ iff for every model $\mathcal{M} \models T$ implies $\mathcal{M} \models \phi$. Let $\mathcal{D}_g(T) = \{\mathbf{O} : \tau \mid \tau \in T\}$.

$$I \text{ have a winning strategy for } \mathbf{DG}^{\mathcal{D}_g(T)}(\mathbf{P} : \phi, \rho) \iff T \models_g \phi.$$

4. **A game for validity over frames:** let $\mathfrak{F}$ be a class of frames characterized by the set of formulas F .

$$I \text{ have a winning strategy for } \mathbf{DG}^{\mathcal{D}_g(F)}(\mathbf{P} : \phi, \rho) \iff \models_{\mathfrak{F}} \phi.$$

5. **A game for entailment over frames:** let $\mathbf{EF}(F, T, \phi, \rho)$ be the following game: in the first round, *You* choose a nominal i . The game then continues with $\mathbf{DG}^{\mathcal{D}_g(F) \cup \mathcal{D}_i(T)}(\mathbf{P}, i : \phi, \rho)$

$$I \text{ have a winning strategy for } \mathbf{EF}(F, T, \phi, \rho) \iff T \models_{\mathfrak{F}} \phi.$$

The proofs readily follow from the previous results. For example, let us consider the game for entailment. Suppose, *I* have a winning strategy for $\mathbf{E}(T, \phi, \rho)$. Let $\mathcal{M}$ be a model and w a world, and let i be a name of w . By assumption, *I* have a winning strategy for $\mathbf{DG}^{\mathcal{D}_i(T)}(\mathbf{P}, i : \phi)$, since *You* may choose i in the first round. By Theorem 9, *I* have a winning strategy for $\mathbf{DG}^{\mathcal{D}_i(T)}(\mathcal{M}, \mathbf{P}, i : \phi, \rho)$. By Proposition 7, *I* have a winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, i : \phi)$ or for $\mathbf{G}(\mathcal{M}, \mathbf{O}, i : \tau)$ for some $\tau \in T$. In the first case, $\mathcal{M}, g(i) \models \phi$ and in the second case $\mathcal{M}, g(i) \not\models T$, both by Theorem 5. If, on the other hand, *You* have a winning strategy for $\mathbf{E}(T, \phi, \rho)$, then for some i , *You* must have a winning strategy for $\mathbf{DG}^{\mathcal{D}_i(T)}(\mathbf{P}, i : \phi)$. Hence, by Theorem 10 and Proposition 7, there is a model $\mathcal{M}$ such that *You* have winning strategies for $\mathbf{G}(\mathcal{M}, \mathbf{P}, i : \phi)$ and for all $\mathbf{G}(\mathcal{M}, \mathbf{O}, i : \tau)$, where $\tau \in T$. By Theorem 5, $\mathcal{M}, g(i) \models T$ and $\mathcal{M}, g(i) \not\models \phi$.

EXAMPLE 7.

The set $F = \{R(i, j) \wedge R(j, k) \rightarrow R(i, k) \mid i, j, k \in N\}$ characterizes the class of transitive frames. Let i be *Your* choice in the first round of $\mathbf{DG}^{\mathcal{D}_g(F)}(\mathbf{P} : \Diamond\Diamond p \rightarrow \Diamond p, \rho)$. The rest of the winning strategy is depicted in Figure 7, where again j and k are *Your* choices for $\Diamond\Diamond p$ and $\Diamond p$, respectively. This winning strategy can be seen as a proof that $\Diamond\Diamond p \rightarrow \Diamond p$ is valid over transitive frames.

7 From strategies to proofs

7.1 Your optimal choices

In this section, we want to modify the disjunctive game so that it becomes finitely branching in ‘Y’-nodes. This alteration will help us conveniently formulate the disjunctive game as a calculus. Infinite branching occurs only in the case of the rules (R_U) , $(R_\Box)$ and $(R_\Diamond)$, where branching is parametrized by the nominals. We will show that in these situations, there is an optimal choice for *You*, so *I* can expect *You* to play according to this choice.

First, we need to define substitutions formally. For a sequence x , let x_n denote its n th element. Let ϕ be a formula and a and b two sequences of nominals of the same length, where every nominal occurs only once in each sequence. We define $\phi[a/b]$ as the formula obtained by simultaneously substituting for every number n all occurrences of a_n in ϕ with b_n . For example, let $a = \langle i, j \rangle$, $b = \langle k, l \rangle$ and $\phi = @_i(k \vee j)$. Then $\phi[a/b] = @_k(k \vee l)$. Note that a substitution also happened in the index of the $@$ -operator. As another example let $a = \langle i_1, i_2, \dots \rangle$ and $b = \langle i_2, i_3, \dots \rangle$. Then $i_1[a/b] = i_2$, because substitution happens simultaneously. We say that ψ is a *pure instance* of ϕ if $\psi = \phi[a/b]$ for some a, b .

We need to extend the notion of substitution to game states: for a game state $g = \mathbf{Q}, i : \phi$ of the semantic game and two sequences of nominals a, b , we define the substitution $g[a/b]$ as $\mathbf{Q}, i[a/b] : \phi[a/b]$. For a set S (of game states or formulas) let $S[a/b] = \{s[a/b] : s \in S\}$. Call a nominal j an *eigenvariable* for S , if it is closed under substitution for j , i.e. $S[j/j'] \subseteq S$ for every nominal j' . We call a set of game states $\mathcal{D}$ *compact* if there are infinitely many eigenvariables for $\mathcal{D}$.

PROPOSITION 15.

Let ρ be fair, $\mathcal{D}$ be compact and let j be an eigenvariable for $\mathcal{D}$ not occurring in D or ϕ . Then

1. *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D \vee \mathbf{P} : \phi, \rho)$ iff *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D \vee \mathbf{P}, j : \phi, \rho)$.
2. *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D \vee \mathbf{P}, i : \Box\phi, \rho)$ iff *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D \vee \mathbf{P}, j : \neg R(i, j) \vee \phi, \rho)$.
3. *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D \vee \mathbf{O}, i : \Diamond\phi, \rho)$ iff *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D \vee \mathbf{P}, j : R(i, j) \wedge \phi, \rho)$.

This result implies that *My* winning strategies in the disjunctive game can be finitely represented: in every disjunctive state whose children branch over the nominals, it is enough to consider a single child only, given by an eigenvariable j not appearing in that disjunctive state. The following result shows that the same can be achieved for the games for entailment and entailment over frames, where *You* choose a nominal in the first round:

PROPOSITION 16.

Let ρ be a fair regulation, T be compact and i be an eigenvariable for T .

1. *You* have a winning strategy in $\mathbf{E}(T, \phi, \rho)$ iff *You* have a winning strategy for $\mathbf{DG}^{\mathcal{D}_i(T)}(\mathbf{P}, i : \phi)$.
2. *You* have a winning strategy in $\mathbf{EF}(F, T, \phi, \rho)$ iff *You* have a winning strategy for $\mathbf{DG}^{\mathcal{D}_g(F) \cup \mathcal{D}_i(T)}(\mathbf{P}, i : \phi)$.

EXAMPLE 8.

Continuing Example 7, remember we had $\mathcal{D} = \{\mathbf{O}, i : R(i, j) \wedge R(j, k) \rightarrow R(i, k) \mid i, j, k \in N\}$, which is closed under all pure instances. Therefore, we see that the choices of i (in the game's first move starting in $\mathbf{P} : \Diamond\Diamond p \rightarrow \Diamond p$), j and k are optimal for *You* in Figure 7. Therefore, this finite winning strategy should be accepted as winning strategy for the infinitely branching game $\mathbf{DG}^{\mathcal{D}}(\mathbf{P} : \Diamond\Diamond p \rightarrow \Diamond p, \rho)$.

The rest of the subsection is devoted to proving Proposition 15. The proof is rather technical and split up into a series of smaller lemmas. For a model $\mathcal{M}$, let $\mathcal{M}[a/b]$ be the same as $\mathcal{M}$, except for the denotation function: $g_{[a/b]}(i) = g(i[a/b])$.

LEMMA 17.

Let $\mathcal{M}[a/b]$ be named. Then $g_{[a/b]}$ restricted to $N[b/a]$ is surjective.

Proof. For a sequence x , let $\text{range}(x)$ denote the set of its elements. We have

$$N[b/a] = N \setminus \text{range}(b) \cup \text{range}(a) = N \setminus (\text{range}(b) \setminus \text{range}(a)).$$

Since $\mathcal{M}[a/b]$ is named, $g_{[a/b]} : N \rightarrow W$ is surjective. It is therefore enough to show that every world with $g_{[a/b]}$ -name in $N \setminus N[b/a]$ also has a $g_{[a/b]}$ -name in $N[b/a]$. So let $w = g_{[a/b]}(i)$ with $i \in N \setminus N[b/a] = \text{range}(b) \setminus \text{range}(a)$, i.e. i is of the form b_m . But then a_m is a $g_{[a/b]}$ -name of w in $N[b/a]$:

$$g_{[a/b]}(a_m) = g(a_m[a/b]) = g(b_m) = g(b_m[a/b]) = g_{[a/b]}(b_m) = g_{[a/b]}(i),$$

where the third equality follows from the fact that $b_m \notin \text{range}(a)$. □

LEMMA 18.

Let a, b be sequences of nominals, such that $\text{range}(a) \subseteq \text{range}(b)$, let $\mathcal{M}[a/b]$ be named. If *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, g)$, then *You* have a winning strategy for $\mathbf{G}(\mathcal{M}[a/b], g[b/a])$.

Proof. Let σ be a winning strategy for *You* for $\mathbf{G}(\mathcal{M}, g)$. By Proposition 6 and Lemma 17 and the assumption that $\mathcal{M}[b/a]$ is named, it is enough to show that $\sigma[b/a]$ is a winning strategy in the semantic game, where branching over the nominals in the rules $(R_{\Box})$, $(R_{\Diamond})$ and (R_U) is restricted to $N[b/a]$.

We must show that $\sigma[b/a]$ is indeed a well-formed winning strategy, i.e. (a) if $h \in \sigma[b/a]$ is labelled 'Y', then $\sigma[b/a]$ contains exactly one child of h and (b) if $h \in \sigma[b/a]$ is labelled 'I', then $\sigma[b/a]$ contains all children of h and (c) all elementary game states in $\sigma[b/a]$ are winning for *You*. By assumption on σ , it is enough to show (a) if g' is a child of g , then $g'[b/a]$ is a child of $g[b/a]$, (b) every child of $g[b/a]$ is of the form $g'[b/a]$, where g' is a child of g , (c) g over $\mathcal{M}$ has the same label as $g[b/a]$ over $\mathcal{M}[a/b]$.

The proof of (a) is straightforward and uses the fact that substitution keeps the syntactic structure of a game state intact. For example, $\mathbf{P}, i[b/a] : \phi[b/a] = (\mathbf{P}, i : \phi)[b/a]$ is a child of $(\mathbf{P}, i : \phi \wedge \psi)[b/a] = \mathbf{P} : i[b/a] : \phi[b/a] \wedge \psi[b/a]$.

The proof of (b) is similar for the logical connectives. In the case of the rules $(R_{\Box})$, $(R_{\Diamond})$ and (R_U) , however, we encounter the following problem. Take, for example, the game state $\mathbf{O} : \phi$. The set of its children is $C = \{\mathbf{O}, i : \phi \mid i \in N\}$, but

$$C[b/a] = \{\mathbf{O}, i[b/a] : \phi[b/a] \mid i \in N\} = \{\mathbf{O}, j : \phi[b/a] \mid j \in N[b/a]\},$$

i.e. the children $\mathbf{O}, j : \phi[b/a]$ of $(\mathbf{O} : \phi)[b/a]$, where $i \in N \setminus N[b/a]$ are not contained in $\sigma[b/a]$. This is where we use the fact from the first paragraph, i.e. it is enough if $\sigma[b/a]$ is a winning strategy for the reformulation of the game with branching restricted to $N[b/a]$.

For (c), the following identity comes in handy: $i[b/a][a/b] = i$ for all nominals i , as it implies $g[b/a](i[b/a]) = g(i)$. For its proof, notice that if $i \notin \text{range}(b)$, then $i \notin \text{range}(a)$, hence $i[b/a][a/b] = i[a/b] = i$. If $i \in \text{range}(b)$, then $i = b_m$ for some m and hence $i[b/a][a/b] = b_m[b/a][a/b] = a_m[a/b] = b_m = i$.

If now g is of the form $\mathbf{O}, i : j$ then it has label ‘Y’ if and only if $g(i) = g(j)$. By the above identity, this is the case iff $g[a/b](i[b/a]) = g[a/b](j[b/a])$, which is equivalent to $\mathbf{O}, i[b/a] : j[b/a]$ being labelled ‘Y’ over $\mathcal{M}[a/b]$. The other cases are similar. $\square$

The proof of the following lemma is simpler than the previous one and uses Proposition 17 in a similar way. We therefore omit its proof.

LEMMA 19.

If $g(j) = g(k)$ and *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, g)$, then *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, g[j/k])$.

Proof of Proposition 15. We will show 2. The direction from right to left is clear, so let us assume *You* have a winning strategy in $\mathbf{DG}^{\mathcal{D}}(D \vee \mathbf{P}, i : \Box\phi, \rho)$ with j as in the assumption. By Theorem 11 and Proposition 8, there is a named model $\mathcal{M}$ such that *You* have winning strategies for $\mathbf{G}(\mathcal{M}, g)$ for all $g \in \mathcal{D} \cup D$ and for $g = \mathbf{P}, i : \Box\phi$. The latter implies that *You* have a winning strategy for $\mathbf{P}, k : \neg R(i, k) \vee \phi$ for some nominal k .

By compactness of $\mathcal{D}$, there is a sequence of distinct eigenvariables $j_1, j_2, \dots$ not occurring in D or ϕ and different from k and j . Let $a = \langle j, j_1, j_2, \dots \rangle$ and $b = \langle k, j, j_1, j_2, \dots \rangle$. Clearly $\mathcal{M}[a/b]$ is named, and $\text{range}(a) \subseteq \text{range}(b)$, therefore Lemma 18 applies and gives *You* winning strategies

- $\mathbf{G}(\mathcal{M}[a/b], \mathbf{P}, j : R(i, j) \vee \phi)$ from *Your* winning strategy for $\mathbf{G}(\mathcal{M}, \mathbf{P}, k : \neg R(i, k) \vee \phi)$.
- $\mathbf{G}(\mathcal{M}[a/b], g)$, where $g \in D$: First, by the lemma, *You* obtain a winning strategy for $\mathbf{G}(\mathcal{M}[a/b], g[b/a])$ from *Your* winning strategy for $\mathbf{G}(\mathcal{M}, g)$. Since $j, j_1, \dots$ do not appear in D , we have that $g[b/a] = g[k/j]$. By Lemma 19, *You* have a winning strategy for $g[k/j][j/k] = g$, since $g[a/b](j) = g[a/b](k)$.
- $\mathbf{G}(\mathcal{M}[a/b], g)$ for $g \in \mathcal{D}$: Since $\mathcal{D}$ is closed under substitution by eigenvariables, *You* have a winning strategy for $\mathbf{G}(\mathcal{M}, g[a/b])$. By the lemma, *You* have a winning strategy for $\mathbf{G}(\mathcal{M}[a/b], g[k/j])$, as $g[a/b][b/a] = g[k/j]$. By Lemma 19, *You* have a winning strategy for $g[k/j][j/k] = g$. $\square$

7.2 The proof system $\mathbf{DS}(\mathcal{D})$

We are now ready to formulate the sequent calculus $\mathbf{DS}(\mathcal{D})$ (Figure 9). We say that a string $i : \phi$ consisting of a hybrid logic formula ϕ and a nominal i is a *labelled formula*. We call an object $\Gamma \Rightarrow \Delta$, where Γ and Δ are multisets of formulas and labelled formulas, a *sequent*. A disjunctive state D can be rewritten as a sequent $\Gamma \Rightarrow \Delta$ in the following way: Γ comprises all (labelled) formulas with the prefix $\mathbf{O}$ in D and Δ comprises all (labelled) formulas with the prefix $\mathbf{P}$ in D . For example, the disjunctive game state $\mathbf{O}, i : \Box p \vee \mathbf{P} : p$ becomes the sequent $i : \Box p \vdash p$. There is thus a 1-1 correspondence between finite sequents and disjunctive states. We will therefore use the notation for disjunctive game states and sequents interchangeably.

Apart from the encoding of disjunctive states as sequents and the traditional bottom-up notation of proof trees, proofs in $\mathbf{DS}(\mathcal{D})$ exactly correspond to *My* winning strategies in the disjunctive game:

Initial sequents

$\Gamma \Rightarrow \Delta$, where $D(\Gamma \Rightarrow \Delta)$ is elementary and winning

Structural Rules

$$\frac{\Gamma, i : \phi, i : \phi \Rightarrow \Delta}{\Gamma, i : \phi \Rightarrow \Delta} (L_c) \qquad \frac{\Gamma \Rightarrow i : \phi, i : \phi, \Delta}{\Gamma \Rightarrow i : \phi, \Delta} (R_c)$$

Context rules ($\bar{\phi}$ is a (possibly labelled) formula)

$$\frac{\Gamma, \bar{\phi} \Rightarrow \Delta}{\Gamma \Rightarrow \Delta} (L_D) \qquad \frac{\Gamma \Rightarrow \bar{\phi}, \Delta}{\Gamma \Rightarrow \Delta} (R_D)$$

(if $\bar{\phi} \in \Sigma$) (if $\bar{\phi} \in \Pi$)

Propositional rules

$$\frac{\Gamma, i : \phi \Rightarrow \Delta \quad \Gamma, i : \psi \Rightarrow \Delta}{\Gamma, i : \phi \vee \psi \Rightarrow \Delta} (L_\vee) \qquad \frac{\Gamma \Rightarrow i : \phi, \Delta}{\Gamma \Rightarrow i : \phi \vee \psi, \Delta} (R_\vee^1)$$

$$\frac{\Gamma, i : \phi \Rightarrow \Delta}{\Gamma, i : \phi \wedge \psi \Rightarrow \Delta} (L_\wedge^1) \qquad \frac{\Gamma \Rightarrow i : \psi, \Delta}{\Gamma \Rightarrow i : \phi \vee \psi, \Delta} (R_\vee^2)$$

$$\frac{\Gamma, i : \psi \Rightarrow \Delta}{\Gamma, i : \phi \wedge \psi \Rightarrow \Delta} (L_\wedge^2) \qquad \frac{\Gamma \Rightarrow i : \phi, \Delta \quad \Gamma \Rightarrow i : \psi, \Delta}{\Gamma \Rightarrow i : \phi \wedge \psi, \Delta} (R_\wedge)$$

$$\frac{\Gamma \Rightarrow i : \phi, \Delta \quad \Gamma, i : \psi \Rightarrow \Delta}{\Gamma, i : \phi \rightarrow \psi \Rightarrow \Delta} (L_\rightarrow) \qquad \frac{\Gamma, i : \phi \Rightarrow \Delta}{\Gamma \Rightarrow i : \phi \rightarrow \psi, \Delta} (R_\rightarrow^1)$$

$$\frac{\Gamma \Rightarrow i : \phi, \Delta}{\Gamma, i : \neg \phi \Rightarrow \Delta} (L_\neg) \qquad \frac{\Gamma \Rightarrow i : \psi, \Delta}{\Gamma \Rightarrow i : \phi \rightarrow \psi, \Delta} (R_\rightarrow^2)$$

$$\frac{\Gamma, i : \phi \Rightarrow \Delta}{\Gamma \Rightarrow i : \neg \phi, \Delta} (R_\neg)$$

Modal rules

$$\frac{\Gamma, i : \phi \vdash \Delta}{\Gamma, \phi \vdash \Delta} (L_U) \qquad \frac{\Gamma \Rightarrow i : \phi, \Delta}{\Gamma \Rightarrow \phi, \Delta} (R_U)$$

$$\frac{\Gamma, j : \neg R(i, j) \vee \phi \Rightarrow \Delta}{\Gamma, i : \Box \phi \Rightarrow \Delta} (L_\Box) \qquad \frac{\Gamma \Rightarrow j : \neg R(i, j) \vee \phi, \Delta}{\Gamma \Rightarrow i : \Box \phi, \Delta} (R_\Box)$$

$$\frac{\Gamma, j : R(i, j) \wedge \phi \Rightarrow \Delta}{\Gamma, i : \Diamond \phi \Rightarrow \Delta} (L_\Diamond) \qquad \frac{\Gamma \Rightarrow j : R(i, j) \wedge \phi, \Delta}{\Gamma \Rightarrow i : \Diamond \phi, \Delta} (R_\Diamond)$$

$$\frac{\Gamma, j : \phi \Rightarrow \Delta}{\Gamma, i : @_j \phi \Rightarrow \Delta} (L_@) \qquad \frac{\Gamma \Rightarrow j : \phi, \Delta}{\Gamma \Rightarrow i : @_j \phi, \Delta} (R_@)$$

In the rules $(R_\Box)$, $(L_\Diamond)$ and (R_U) , the nominal j is an eigenvariable for $\Sigma \Rightarrow \Pi$ not appearing in the lower sequent. This condition corresponds to *Your* optimal choices.

FIGURE 9. The proof system $\mathbf{DS}(\Sigma \Rightarrow \Pi)$.

the order of rule application defines a regulation function and determines *My* moves in the strategy, branching in the proof tree corresponds to branching in the winning strategy, i.e. *Your* possible moves. Infinitary branching is modified according to the discussion in the previous subsection. Duplication in the game takes the form of left and right contraction rules. The base rules are exactly

the (encoding¹³ of) winning disjunctive elementary states and thus winning for *Me*. Let us write $\Sigma \Rightarrow \Pi \vdash \Gamma \Rightarrow \Delta$ if $\Gamma \Rightarrow \Delta$ is provable in **DS**($\Sigma \Rightarrow \Pi$).¹⁴ We write $\Sigma \vdash \Delta$ if $\Pi = \emptyset$ and $\Gamma = \emptyset$. Using this correspondence we immediately get the fact that the disjunctive game is adequate with respect to the calculus, as well as the following reformulations of the results in Section 6:

THEOREM 20.

Let ρ be fair and $\Sigma \Rightarrow \Pi$ compact. Then $\Sigma \Rightarrow \Pi \vdash \Gamma \Rightarrow \Delta$ iff I have a winning strategy for $\mathbf{DG}^{\Sigma \Rightarrow \Delta}(\Gamma \Rightarrow \Delta, \rho)$.

COROLLARY 21.

Let ϕ be a formula, T a compact set of formulas and $\mathfrak{F}$ a class of frames characterized by F .

1. $\vdash \phi$ iff ϕ is valid,
2. $\{i : \tau \mid \tau \in T\} \vdash i : \phi$ iff $T \models \phi$,
3. $T \vdash \phi$ iff $T \models_g \phi$,
4. $F \vdash \phi$ iff $\models_{\mathfrak{F}} \phi$,
5. $F \cup \{i : \tau \mid \tau \in T\} \vdash i : \phi$ iff $T \models_{\mathfrak{F}} \phi$.

In 2 and 5, i is an eigenvariable of T not appearing in ϕ .

The calculus **DS** takes a familiar form: the rules for the logical connectives are the (labelled) versions of the usual propositional rules of sequent calculus for classical logic. The modal operator rules come in the form of their first-order translations. Apart from the axioms, **DS** can therefore be seen as a fragment of the usual sequent system for first-order logic. In turn, **DS** is an extension of the sequent calculus **G3K** [16] to hybrid logic. Similarly to **G3K**, a failed proof search in **DS** directly gives rise to a countermodel. This follows from our proof of Theorem 11, where the explicit countermodel $\mathcal{M}_\pi$ was constructed. We will investigate the connections to other calculi in Subsection 7.4.

EXAMPLE 9.

Let F characterize the set of transitive frames, i.e. $F = \{R(i, j) \wedge R(j, k) \rightarrow R(i, k) \mid i, j, k \in N\}$. We show that the following rule scheme for transitivity is admissible in **DS**(F):

$$\frac{\Gamma \Rightarrow R(i, j), \Delta \quad \Gamma \Rightarrow R(j, k), \Delta \quad \Gamma, R(i, k) \Rightarrow \Delta}{\Gamma \Rightarrow \Delta} \text{ (trans)}$$

The following derivation resembles *My* winning strategy in Figure 8:

$$\frac{\frac{\Gamma \Rightarrow R(i, j), \Delta \quad \Gamma \Rightarrow R(j, k), \Delta}{\Gamma \Rightarrow R(i, j) \wedge R(j, k), \Delta} (R_\wedge) \quad \Gamma, R(i, k) \Rightarrow \Delta}{\frac{\Gamma, R(i, j) \wedge R(j, k) \rightarrow R(i, k), \Delta}{\Gamma \Rightarrow \Delta} (L_{\mathcal{D}})} (L_{\rightarrow})$$

This example shows that we can use the game-theoretic approach to formulate structural rules for classes of frames that are characterizable in hybrid language. We could therefore reformulate our proof system with these structural rules in place of the context-rules (to show that **DS** + (*trans*) derives the transitivity-axioms, we can use invertibility of the logical rules), similar to **G3K** [16]. The rule for transitivity there is easily seen to be derivable using (*trans*), and (*trans*) is admissible in **G3K** plus the transitivity rule.

¹³We say that $\Gamma \vdash \Delta$ is *elementary/winning* iff its corresponding disjunctive game state is elementary/winning.

¹⁴Here, Σ and Π could be infinite and we identify this infinite sequent with the set of its corresponding game states.

EXAMPLE 10.

We show how to derive $\Diamond\Diamond p \rightarrow \Diamond p$ in $\mathbf{DS}(F)$, where F characterizes the class of transitive frames, i.e. $F = \{R(i, j) \wedge R(j, k) \rightarrow R(i, k) \mid i, j, k \in N\}$. The following proof is just a rewriting of My winning strategy in the game $\mathbf{DG}^{\mathcal{D}_g(F)}(\mathbf{P} : \Diamond\Diamond p \rightarrow \Diamond p)$ from Example 7. In all derivations, we omit writing down the label of relation claims:

$$\begin{array}{c}
\frac{R(i, j), R(j, k), k : p \Rightarrow R(i, k) \quad R(i, j), R(j, k), k : p \Rightarrow k : p}{R(i, j), R(j, k), k : p \Rightarrow k : R(i, k) \wedge p} (L_{\wedge}) \\
\frac{\quad}{R(i, j), R(j, k), k : p \Rightarrow i : \Diamond p} (R_{\Diamond}) \\
\frac{\quad}{R(i, j), k : R(j, k) \wedge p \Rightarrow i : \Diamond p} (L_{\wedge}) \times 2, (R_c) \times 2 \\
\frac{\quad}{R(i, j), j : \Diamond p \Rightarrow i : \Diamond p} (L_{\Diamond}) \\
\frac{\quad}{j : R(i, j) \wedge \Diamond p \Rightarrow i : \Diamond p} (L_{\wedge}) \times 2, (R_c) \times 2 \\
\frac{\quad}{i : \Diamond\Diamond p \Rightarrow i : \Diamond p} (L_{\Diamond}) \\
\frac{\quad}{\Rightarrow i : \Diamond\Diamond p \rightarrow \Diamond p} (R_{\rightarrow}) \times 2, (R_c) \times 2 \\
\frac{\quad}{\Rightarrow \Diamond\Diamond p \rightarrow \Diamond p} (R_U)
\end{array}$$

The proof at the left topmost sequent continues with an application of the *(trans)*-rule from Example 9 with $\Gamma = \{R(i, j), R(j, k), k : p\}$ and $\Delta = \{R(i, k)\}$.

7.3 PTIME checkability of initial sequents

To establish $\mathbf{DS}$ as a suitable proof system, we must demonstrate that we can effectively check whether given elementary sequents are indeed winning, i.e. initial sequents.¹⁵ As an upper limit, checking elementary sequents must be easier than checking validity in hybrid logic, which is known to be PSPACE-complete. In fact, checking elementary sequents turns out to be in PTIME:

PROPOSITION 22.

Checking whether an elementary disjunctive state D is winning for Me is in PTIME.

The proof of this proposition is the topic of the remaining subsection.

DEFINITION 23.

Let $\mathcal{M} = (W^{\mathcal{M}}, R^{\mathcal{M}}, V^{\mathcal{M}}, g^{\mathcal{M}})$ and $\mathcal{N} = (W^{\mathcal{N}}, R^{\mathcal{N}}, V^{\mathcal{N}}, g^{\mathcal{N}})$ be two models. We call a function $f : W^{\mathcal{M}} \rightarrow W^{\mathcal{N}}$ a homomorphism and write $f : \mathcal{M} \rightarrow \mathcal{N}$ iff

1. $f(g^{\mathcal{M}}(i)) = g^{\mathcal{N}}(i)$, for all nominals i ,
2. $wR^{\mathcal{M}}v$ implies $f(w)R^{\mathcal{N}}f(v)$, for all worlds w, v ,
3. $w \in V^{\mathcal{M}}(p)$ implies $f(w) \in V^{\mathcal{N}}(p)$ for all worlds w and variables p .

Let $\mathcal{E}$ be a set of elementary game states. We call a model $\mathcal{M}$ an $\mathcal{E}$ -countermodel if all states of $\mathcal{E}$ are labelled ‘Y’ in the semantic game over $\mathcal{M}$. The following proposition shows that the model $\mathcal{M}_{\mathcal{E}}$ (see Definition 13) is, in a way, the most general $\mathcal{E}$ -countermodel.

¹⁵ Alternatively, one could restrict initial sequents to trivial instances and extend the proof system by structural rules, cf. Section 7.4.

PROPOSITION 24.

Let $\mathcal{N}$ be an $\mathcal{E}$ -countermodel. Then there is a (unique) homomorphism $f : \mathcal{M}_{\mathcal{E}} \rightarrow \mathcal{N}$. In this case, $\mathcal{M}_{\mathcal{E}}$ is an $\mathcal{E}$ -countermodel, too.

Proof. The only way to define such a homomorphism is by setting $f([i]) = g^{\mathcal{N}}(i)$. To check that this function is well-defined, we observe that $i \approx_{\mathcal{E}} j$ implies $g^{\mathcal{N}}(i) = g^{\mathcal{N}}(j)$: let $i \sim_{\mathcal{E}} j$, then either $\mathbf{O}, i : j$ or $\mathbf{O}, j : i$ appears in $\mathcal{E}$. Since $\mathcal{N}$ is an $\mathcal{E}$ -countermodel, $g^{\mathcal{N}}(i) = g^{\mathcal{N}}(j)$. The claim for $\approx_{\mathcal{E}}$ follows by transitivity, reflexivity and symmetry of equality. To see that f is a homomorphism, let $[i]R^{\mathcal{E}}[j]$. This means, for some $i' \approx i$ and $j' \approx j$ and some nominal k , $\mathbf{O}, k : R(i, j)$ is in $\mathcal{E}$. Since $\mathcal{N}$ is an $\mathcal{E}$ -countermodel, $g^{\mathcal{N}}(i')R^{\mathcal{N}}g^{\mathcal{N}}(j')$. By the above observation, $g^{\mathcal{N}}(i)R^{\mathcal{N}}g^{\mathcal{N}}(j)$. Showing that $[i] \in V^{\mathcal{E}}(p)$ implies $g^{\mathcal{N}}(i) \in V^{\mathcal{N}}(p)$ is similar. It follows that $\mathcal{M}_{\mathcal{E}}$ is an $\mathcal{E}$ -countermodel: by construction, all states in $\mathcal{E}$ of the form $\mathbf{O}, i : \phi$ are labelled ‘Y’ in the semantic game over $\mathcal{M}_{\mathcal{E}}$. If $\mathbf{P}, i : j$ is in $\mathcal{E}$, then $g^{\mathcal{N}}(i) \neq g^{\mathcal{N}}(j)$, since $\mathcal{N}$ is an $\mathcal{E}$ -countermodel. This implies $[i] \neq [j]$ and hence $\mathbf{P}, i : j$ is labelled ‘Y’ in $\mathcal{M}_{\mathcal{E}}$. If $\mathbf{P}, k : R(i, j)$ is in $\mathcal{E}$, then $g^{\mathcal{N}}(i)R^{\mathcal{N}}g^{\mathcal{N}}(j)$ and therefore $[i]R^{\mathcal{E}}[j]$. The case for $\mathbf{P}, i : p$ is similar. $\square$

Proof of Proposition 22. If there is a countermodel of D , then by Proposition 24, $\mathcal{M}_D$ is already a countermodel. Thus, it suffices to construct $\mathcal{M}_D$ and check whether the **P**-part of D is false there, since the **O**-part is false by construction. Although the model $\mathcal{M}_D$ is infinite, it can be finitely represented using the **O**-part of D :

- For every nominal i , create a equivalence class (list) containing only i .
- For every $\mathbf{O}, i : j \in D$, merge the equivalence classes of i and j .
- For every $\mathbf{O}, k : R(i, j) \in D$, draw an R -arrow from $[i]$ to $[j]$.
- For every $\mathbf{O}, i : p \in D$, mark $[i]$ with the label p .

Constructing this representation takes polynomial time. Checking whether the **P**-part of D is labelled ‘Y’ in the semantic game over $\mathcal{M}_D$ can be done in polynomial time too:

- For every $\mathbf{P}, i : j \in D$, check that $j \notin [i]$.
- For every $\mathbf{P}, k : R(i, j) \in D$, check that there is no R -arrow from $[i]$ to $[j]$.
- For every $\mathbf{P}, i : p \in D$, check that $[i]$ does not have the label p .

$\square$

7.4 Connections to existing calculi

Our calculus **DS**¹⁶ is closely related to several existing sequent systems in the literature, like Negri’s labelled system for orthodox modal logic [16]. In [3], Blackburn reformulates his tableau system for hybrid logic as a sequent calculus and in [7], Bräuner does the same with his natural deduction system. In this section we investigate the connections to the latter system **G_H** (see Figure 10), as it is the most similar to **DS**. Most of the discussion, however, also applies to the other systems. We will argue that **DS** and **G_H** are essentially equivalent. The differences are

- In the original system **G_H** in [7], formulas are not labelled but instead prefixed by the @-operator. As this is clearly equivalent, we rephrased the system for convenience of comparison.
- In **G_H**, sequents $\Gamma \Rightarrow \Delta$ consist of sets Γ and Δ , rather than multisets in **DS**. Consequently, while we make use of contraction rules in **DS**, they are ‘built-in’ to **G_H**.

¹⁶In this subsection, we consider **DS** minus the rules (R_U) and (L_U) .

Initial sequents

$$\Gamma, i : \phi \Rightarrow i : \phi, \Delta$$

$$\Gamma, i : \perp \Rightarrow \Delta$$

Propositional rules

$$(L_{\wedge}) \frac{\Gamma, i : \phi, i : \psi \Rightarrow \Delta}{\Gamma, i : \phi \wedge \psi \Rightarrow \Delta}$$

$$(R_{\wedge}) \frac{\Gamma \Rightarrow i : \phi, \Delta \quad \Gamma \Rightarrow i : \psi, \Delta}{\Gamma \Rightarrow i : \phi \wedge \psi, \Delta}$$

$$(L_{\rightarrow}) \frac{\Gamma \Rightarrow i : \phi, \Delta \quad \Gamma, i : \psi \Rightarrow \Delta}{\Gamma, i : \phi \rightarrow \psi \Rightarrow \Delta}$$

$$(R_{\rightarrow}) \frac{\Gamma, i : \phi \Rightarrow i : \psi, \Delta}{\Gamma \Rightarrow i : \phi \rightarrow \psi, \Delta}$$

Modal rules

$$(L_{\Box}) \frac{\Gamma \Rightarrow i : \Diamond j, \Delta \quad \Gamma, j : \phi \Rightarrow \Delta}{\Gamma, i : \Box \phi \Rightarrow \Delta}$$

$$(R_{\Box}) \frac{\Gamma, i : \Diamond j \Rightarrow j : \phi, \Delta}{\Gamma \Rightarrow i : \Box \phi, \Delta}$$

$$(L_{@}) \frac{\Gamma, j : \phi \Rightarrow \Delta}{\Gamma, i : @_j \phi \Rightarrow \Delta}$$

$$(R_{@}) \frac{\Gamma \Rightarrow j : \phi, \Delta}{\Gamma \Rightarrow i : @_j \phi, \Delta}$$

Rules for nominals

$$(Ref) \frac{\Gamma, i : i \Rightarrow \Delta}{\Gamma \Rightarrow \Delta}$$

$$(Nom1) \frac{\Gamma \Rightarrow i : j, \Delta \quad \Gamma \Rightarrow i : \phi, \Delta}{\Gamma \Rightarrow j : \phi, \Delta}$$

$$(Nom2) \frac{\Gamma \Rightarrow i : k, \Delta \quad \Gamma \Rightarrow i : \Diamond j, \Delta \quad \Gamma, k : \Diamond j \Rightarrow \Delta}{\Gamma \Rightarrow \Delta}$$

In the rule $(R_{\Box})$ the nominal j must not occur in the lower sequent, in the rule $(Nom1)$, ϕ must be a propositional variable or a nominal.

FIGURE 10. The proof system $\mathbf{G}_{\mathcal{H}}$.

- In $\mathbf{G}_{\mathcal{H}}$, the connectives $\vee, \neg, \Diamond$ are treated as defined. In $\mathbf{DS}$, relational claims $R(i, j)$ are treated as elementary, in $\mathbf{G}_{\mathcal{H}}$ relational claims are written as complex formulas $@_i \Diamond j$.
- The propositional rules of $\mathbf{G}_{\mathcal{H}}$ are derivable in $\mathbf{DS}$. The propositional rules of $\mathbf{DS}$ can be shown to be admissible in $\mathbf{G}_{\mathcal{H}}$ using admissibility of weakening.
- Replacing ' $R(i, j)$ ' with ' $@_i \Diamond j$ ', the modal rules of $\mathbf{G}_{\mathcal{H}}$ are derivable in $\mathbf{DS}$. Writing ' $@_i \Diamond j$ ' instead of ' $R(i, j)$ ', the modal rules of $\mathbf{DS}$ can be shown to be admissible in $\mathbf{G}_{\mathcal{H}}$ using the invertibility of the propositional rules.
- Initial sequents are more complex in $\mathbf{DS}$: there, we allow all valid elementary sequents as axioms. In $\mathbf{G}_{\mathcal{H}}$, there are only 'trivial' axioms, but extra rules (Ref) , $(Nom1)$ and $(Nom2)$.

The last point is perhaps the most interesting one: by completeness of $\mathbf{G}_{\mathcal{H}}$, all the initial sequents in $\mathbf{DS}$ can be derived in $\mathbf{G}_{\mathcal{H}}$ from trivial axioms by using the rules (Ref) , $(Nom1)$, $(Nom2)$ and the diamond-rules restricted to instances where ϕ is a nominal or a propositional variable. Putting things together, we have the following correspondence:

PROPOSITION 25.

1. Let Γ and Δ be finite multisets. Then $\vdash_{\mathbf{DS}} \Gamma \Rightarrow \Delta$ implies $\vdash_{\mathbf{G}_{\mathcal{H}}} \Gamma' \Rightarrow \Delta'$, where Γ' and Δ' are the carrier sets of Γ and Δ respectively and every subformula $R(i, j)$ is replaced by $@_i \Diamond j$.
2. Let Γ and Δ be finite sets. Then $\vdash_{\mathbf{G}_{\mathcal{H}}} \Gamma \Rightarrow \Delta$ implies $\vdash_{\mathbf{DS}} \Gamma' \Rightarrow \Delta'$, where Γ' and Δ' are obtained from Γ and Δ by replacing every subformula $@_i \Diamond j$ by $R(i, j)$ and every labelled formula $i : \Diamond j$ by $j : R(i, j)$.

In particular, this correspondence implies that every proof in $\mathbf{G}_H$ can be structured as follows: starting from the root, apply only propositional rules and modal rules to complex formulas until all leaves are elementary. This is essentially a proof in $\mathbf{DS}$. Then, apply nominal and restricted diamond rules to arrive at trivial axioms.

It must be noted that, in principle, nominal rules can be included into the disjunctive game. For example, the rule corresponding to (*Nom1*) would look as follows: if in the current disjunctive state $D \vee \mathbf{P}, j : \phi$, it is *My* turn, we can allow *Me* to choose a nominal i and then *You* choose whether to continue the game with $D \vee \mathbf{P}, i : j$ or with $D \vee \mathbf{P}, i : \phi$. However, we chose not to include these rules, as they seem to lack a clear motivation in terms of the semantic game. Another way to reformulate the disjunctive game with trivial initial sequents would be to include nominal axioms in the context, for example, all instances of $@_i j \rightarrow @_i i$, $R(i, j) \wedge @_i k \rightarrow R(k, j)$, etc.

8 Conclusion and future work

In this paper, we developed a semantic game for hybrid logic and proved its adequacy: *I* have a winning strategy in the game for ϕ at world w over the model $\mathcal{M}$ iff $\mathcal{M}, w \models \phi$. We modified the disjunctive game [8] to adequately model both validity for hybrid logic and entailment over classes of frames characterizable in hybrid language. We point out that our approach naturally extends to cover other operations common in hybrid logic, like quantification over nominals ($\forall$) and the down-arrow binder ($\downarrow$) [7].

Our aim, however, was not only to give game-theoretic characterizations of truth in a model and entailment, but to do so in a way that allows for a natural lifting of the first to the second: the disjunctive game corresponds to simultaneously playing the semantic game over all possible models. In turn, the search for a winning strategy can be straightforwardly formulated as an analytic calculus.

The lifting of semantic games to analytic calculi via a disjunctive game has been demonstrated on the propositional level for classical logic, Łukasiewicz logic and Gödel logic [8, 9, 17]. Our approach is the first to deal with infinitely many possible actions and to address entailment. We are thus keen to extend the known results to the first-order versions of the above logics. Ultimately, the liftings rely on similar game-theoretic properties of the semantic and the corresponding disjunctive game. We plan to exploit the relevant features of these similarities on an abstract level to design a powerful, general algorithm for lifting game states to disjunctive game states for a wide range of logics.

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